

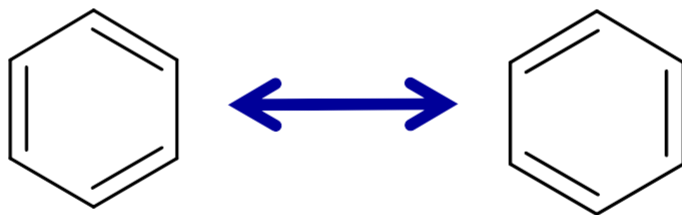
CHAPTER 6
BENZENE &
ITS DERIVATIVES

6.1 INTRODUCTION

- Hydrocarbon that contain rings which are stabilised by **electron delocalisation** are called **aromatic hydrocarbon** or **arenes**.
- Benzene is an example of an aromatic hydrocarbon.

KEKULÉ STRUCTURE

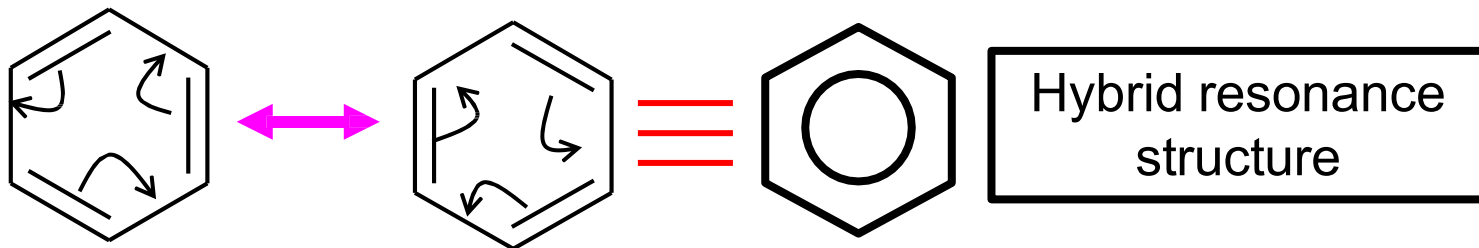
- Kekulé suggested that the carbon atoms of benzene are in the ring, bonded to each other by alternating single and double bonds



- However, Kekule structure can't explain why benzene molecules do not undergo addition reaction.

RESONANCE STRUCTURE

- In 1930s , Linus Pauling introduced the concept of **hybrid orbital and resonance**.
- The real structure of benzene is the resonance hybrid from two Kekulé structure as shown below:



THE CRITERIA FOR AROMATICITY

1

Cyclic.

2

Planar

(all adjacent p-orbital must be aligned)

3

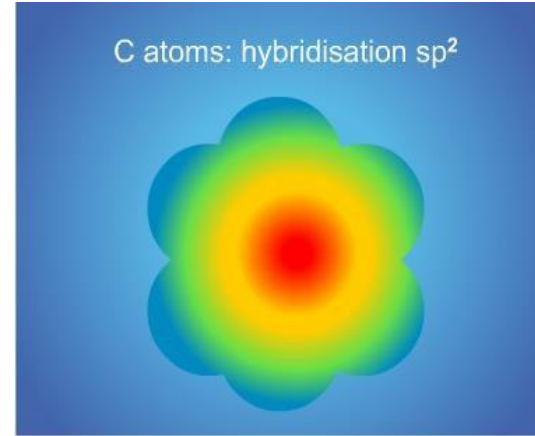
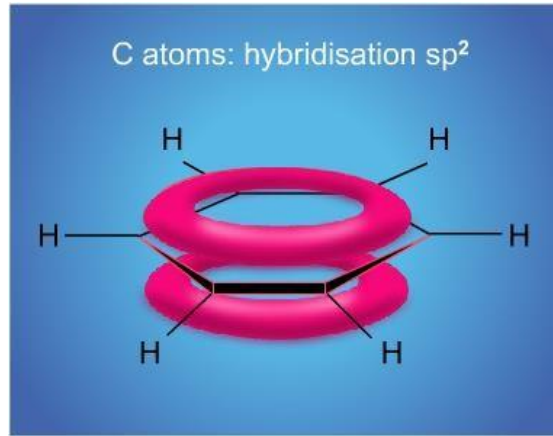
Completely conjugated

(double-bond and single bonds are alternate)

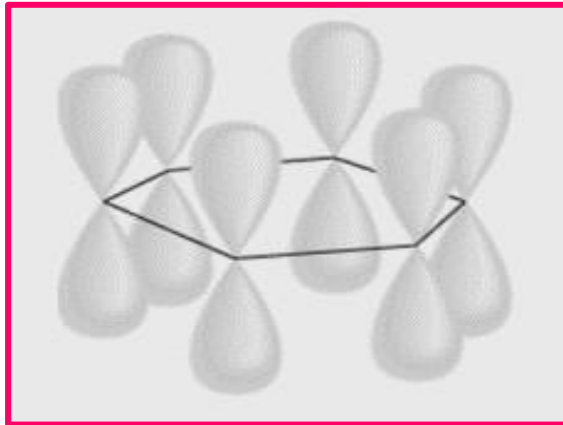
4

Contain particular number of π electrons

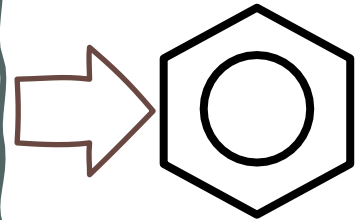
- obeys Hückel's Rule $(4n+2) \pi$ and the electrons are delocalised.



The **six π electrons** completely delocalised around the ring



The **circle** represents the **six π electrons**, distributed over the **six atoms** of the **ring**!



All six C atoms and **six π orbitals** are **equivalent**

THE CRITERIA FOR AROMATICITY

Hückel's Rule



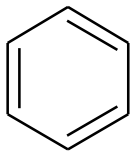
Erich Hückel
(1896-1980)

* cyclic, planar and completely conjugated compounds that contain $[4n+2]$ π electron ($n=0,1,2,\dots$) are said to be aromatic

Planar monocyclic rings with 2, 6, 10, 14 and so forth π electrons are aromatic

THE CRITERIA FOR AROMATICITY

Aromatic compounds
with a **single ring**



number of π electrons = 6

$$4n + 2 = 6\pi$$

$$n = 1$$

Benzene

Benzene is **aromatic** because:

- Contains $(4n+2)\pi$ electrons (obeys Hückels Rule)
- cyclic, planar and completely conjugated

Aromatic compounds
with a **more than one ring**

number of π electrons = 10

$$4n + 2 = 10\pi$$

$$n = 2$$

Naphthalene

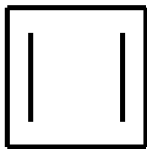
Naphthalene is **aromatic**
because:

- Contains $(4n+2)\pi$ electrons (obeys Hückels Rule)
- cyclic, planar and completely conjugated

EXERCISE

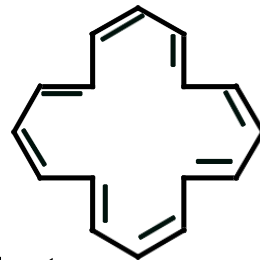
Determine whether these below compounds are aromatic or not.

(a)



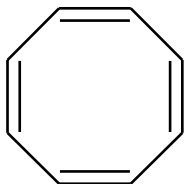
Not aromatic
(do not obeys HÜckels Rule)

(b)



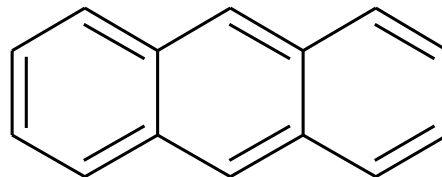
Not aromatic
(do not obeys HÜckels Rule)

(c)



Not aromatic
(do not obeys HÜckels Rule)

(d)



Aromatic

6.2 NOMENCLATURE

1 Monosubstituted benzene

- Benzene as the parent name
- The benzene ring and the substituent combine to form a parent name.

2 disubstituted benzene

- Two same substituents
- Two different substituents

3 Tri and tetrasubstituted benzene

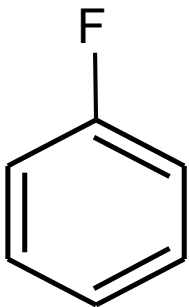
4 Benzene as substituent

Monosubstituted benzene

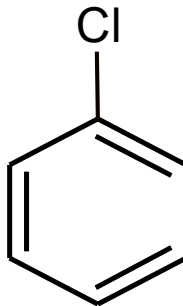
a

Benzene as the parent name

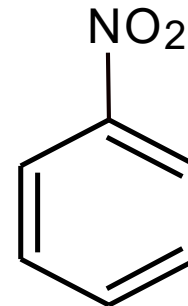
Substituent is indicated by a prefix.



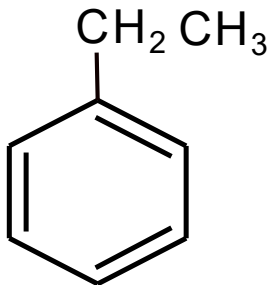
fluorobenzene



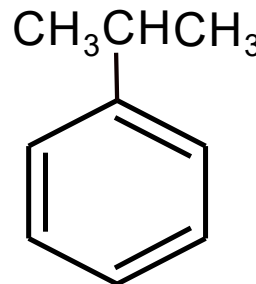
chlorobenzene



nitrobenzene



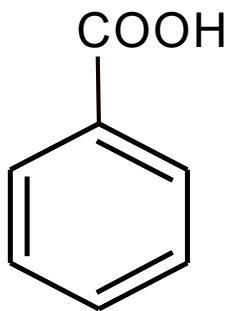
ethylbenzene



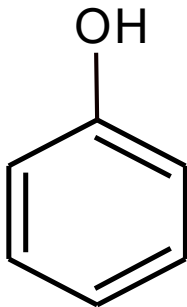
isopropylbenzene

1 Monosubstituted benzene

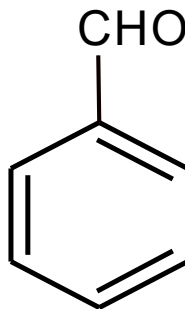
b The benzene ring and the substituent combine to form a parent name.



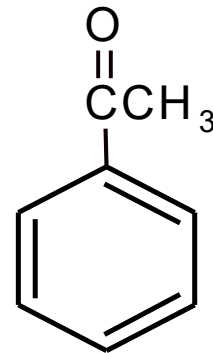
Benzoic acid



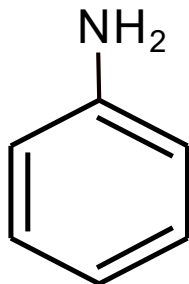
phenol



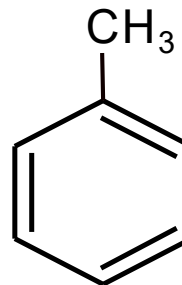
benzaldehyde



acetophenone @
phenylethanone



aniline



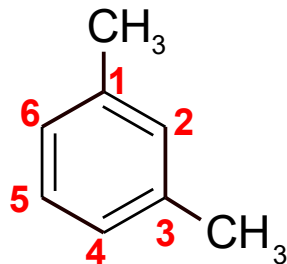
toluene @ methylbenzene

2

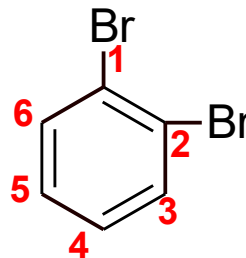
disubstituted benzene

Using numbers or prefixes ortho (1,2), meta(1,3) & para(1,4) to indicate their position.

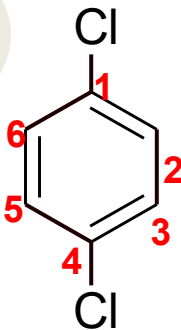
a Two same substituents



1,3-dimethylbenzene @
m-dimethylbenzene



1,2-dibromobenzene @
o-dibromobenzene



1,4-dichlorobenzene @
p-dichlorobenzene

2

disubstituted benzene

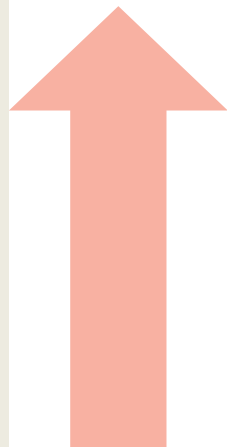
b

Two different substituents

When **more than 1** functional groups present:

- The **highest priority** will be the parent chain and **labelled number 1**.

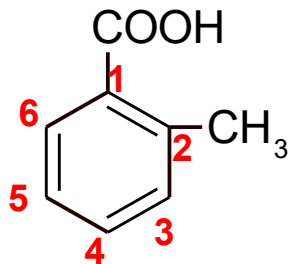
COOH
CHO
C=O
-OH
-NH₂
C=C
C≡C
-R , -NO₂ , -OR, -X



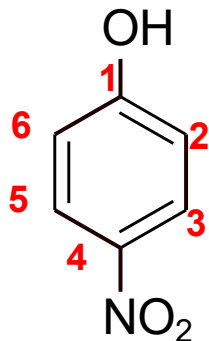
Increasing priority of
functional groups in
nomenclature

2 disubstituted benzene

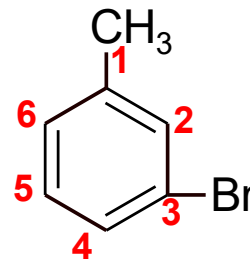
b Two different substituents



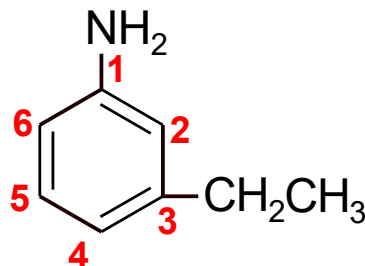
2-methylbenzoic acid @
o-methylbenzoic acid



4-nitrophenol @
p-nitrophenol



3-bromotoluene @
m-bromotoluene

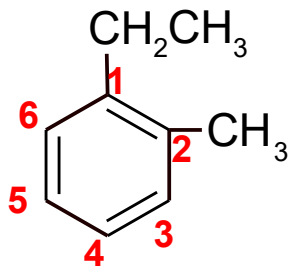


3-ethylaniline @
m-ethylaniline

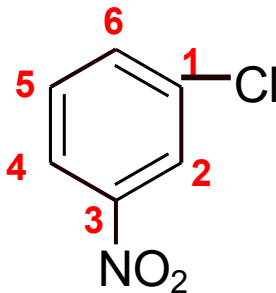
2

disubstituted benzene

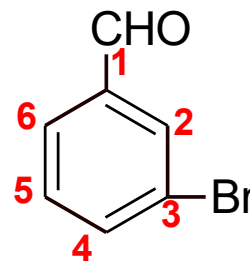
b Two different substituents



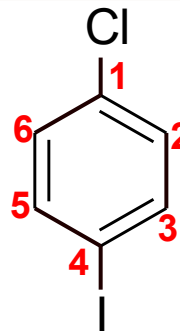
1-ethyl-2-methylbenzene



1-chloro-3-nitrobenzene



3-bromobenzaldehyde

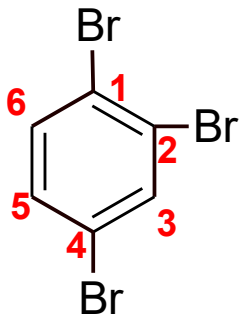


1-chloro-4-iodobenzene

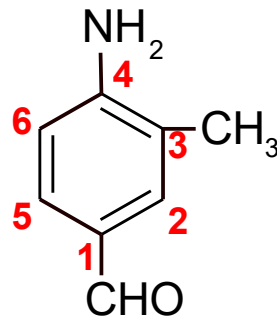
3

Tri and tetrasubstituted benzene

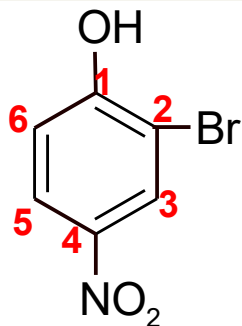
- Using **numbers** to indicate their position.
- The substituents are listed **alphabetically** when writing the name.



1,2,4-tribromobenzene



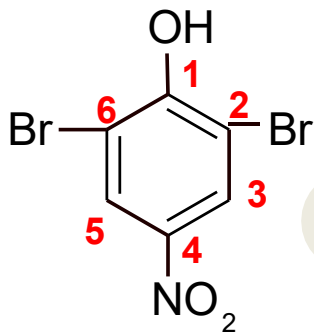
4-amino-3-methylbenzaldehyde



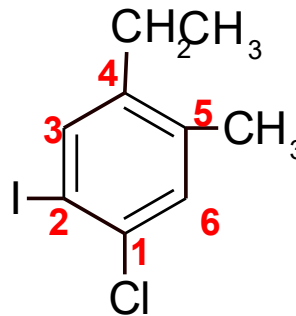
2-bromo-4-nitrophenol

3 Tri and tetrasubstituted benzene

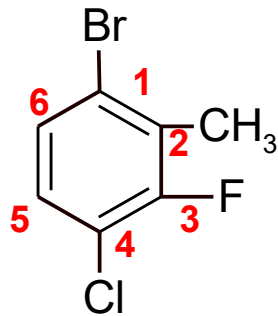
- Using **numbers** to indicate their position.
- The substituents are listed **alphabetically** when writing the name.



2,6-dibromo-4-nitrophenol



1-chloro-2-iodo-4-ethyl-5-methylbenzene

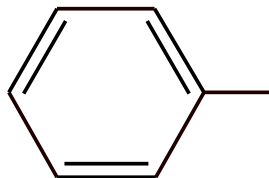


1-bromo-4-chloro-3-flouro-2-methylbenzene

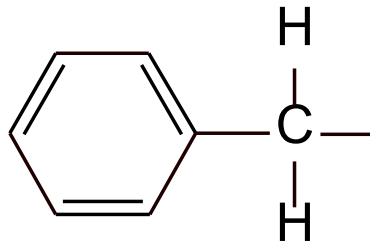
4

Benzene as substituent

- The benzene ring is name as substituent on a parent chain if :
 - alkyl substituent is larger than the ring (more than 6 C).
 - carbon chain containing other functional group
- The substituents are listed **alphabetically** when writing the name.



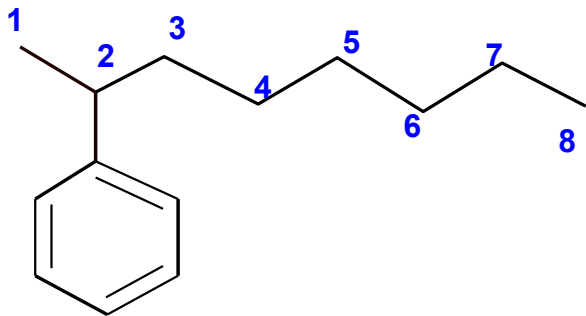
C_6H_5 -
phenyl



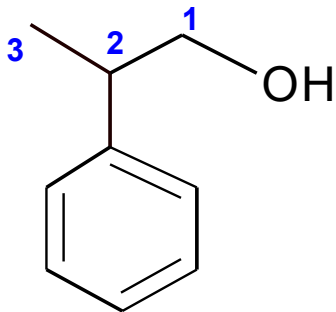
$\text{C}_6\text{H}_5\text{CH}_2$ -
benzyl

4

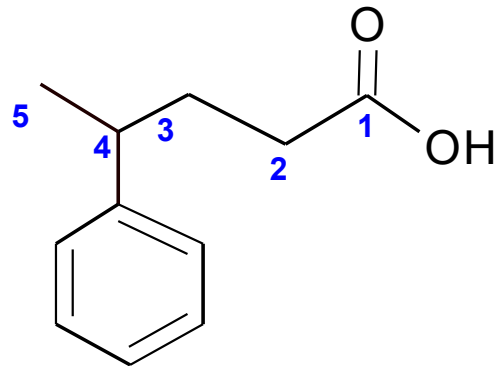
Benzene as substituent



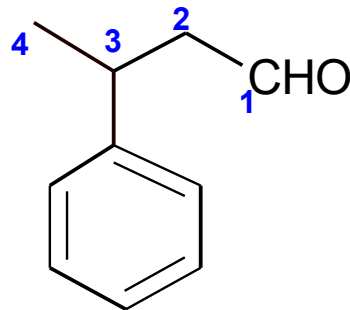
2-phenyloctane



2-phenyl-1-propanol



4-phenylpentanoic acid

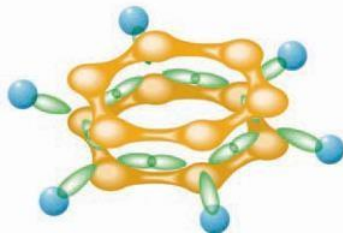
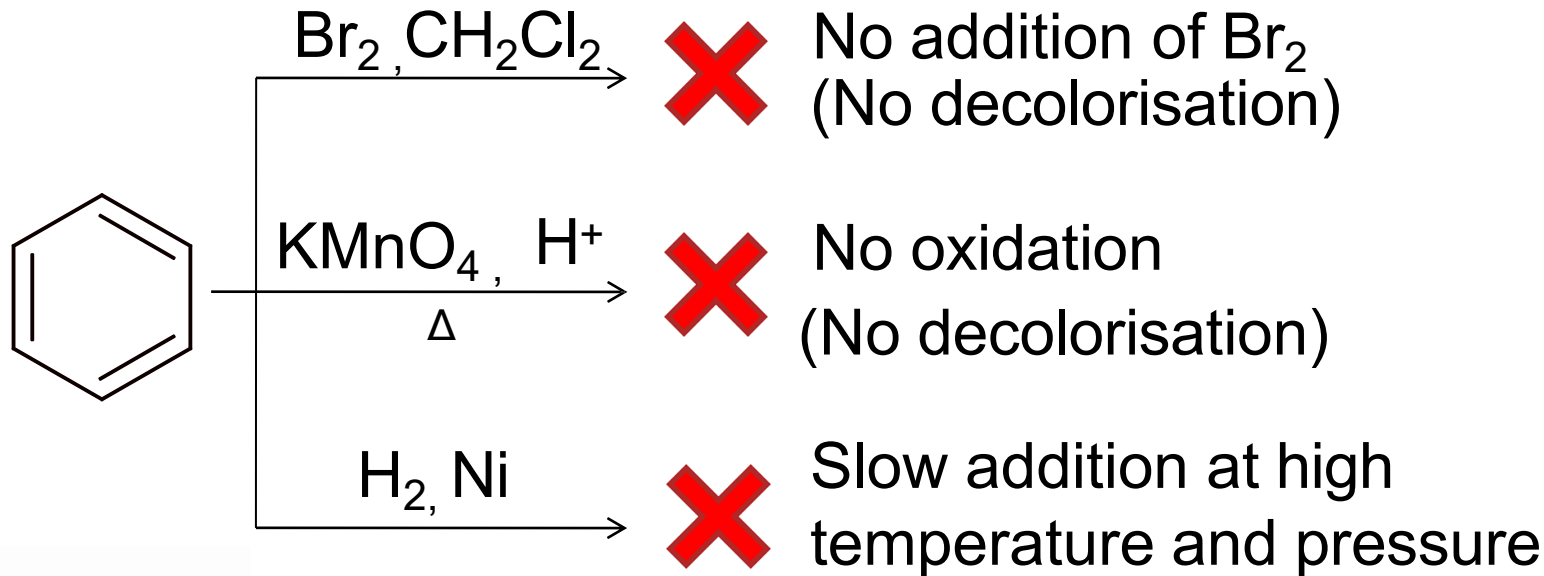


3-phenylbutanal

6.3 CHEMICAL PROPERTIES OF BENZENE

21

UNUSUAL REACTIONS OF BENZENE

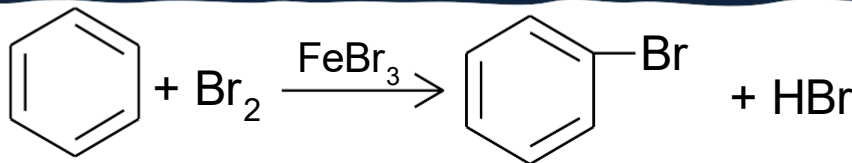


Alkenes readily undergo addition reaction, **benzene does not!**

REACTIONS OF BENZENE

- 1 **Electrophilic aromatic substitution** : Involves the benzene ring itself

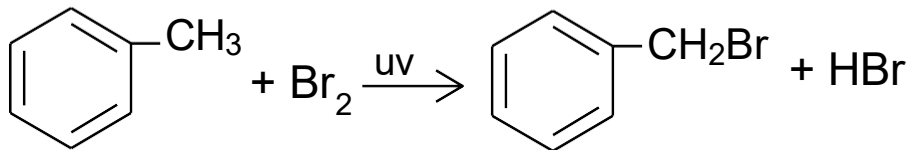
Example:



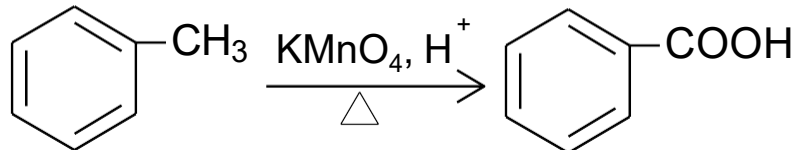
- 2 Reaction involves substituent attached to benzene ring

Example:

free radical substitution



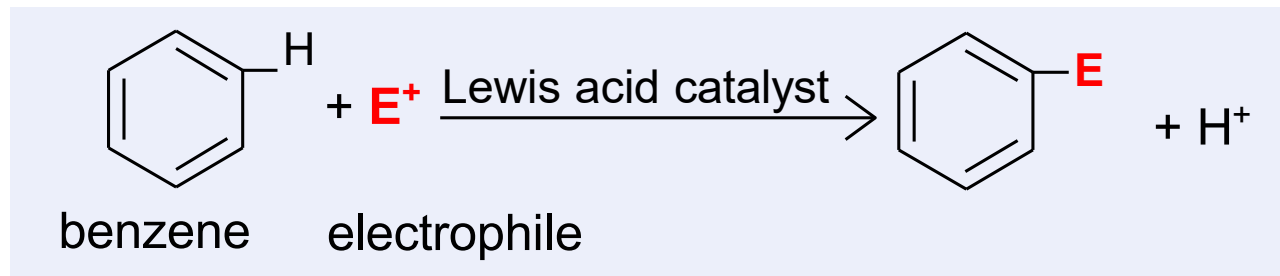
oxidation



1 Electrophilic Aromatic Substitution

1. **Electrophilic Aromatic Substitution** reaction is the characteristic reaction of benzene.
2. In electrophilic substitution reaction, a **hydrogen** atom is **replaced** by an **electrophile**.

General Equation:



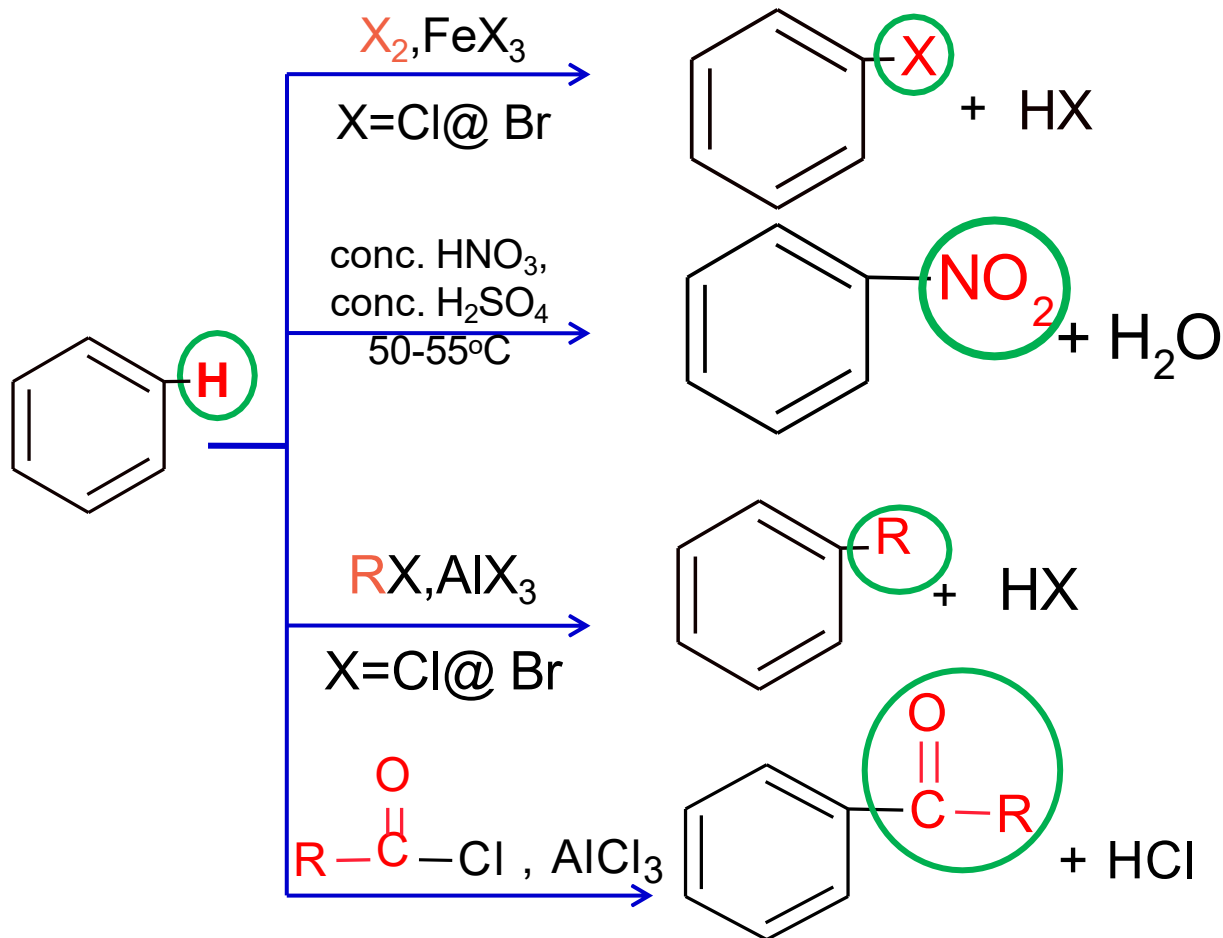
1 Electrophilic Aromatic Substitution

1 Halogenation

2 Nitration

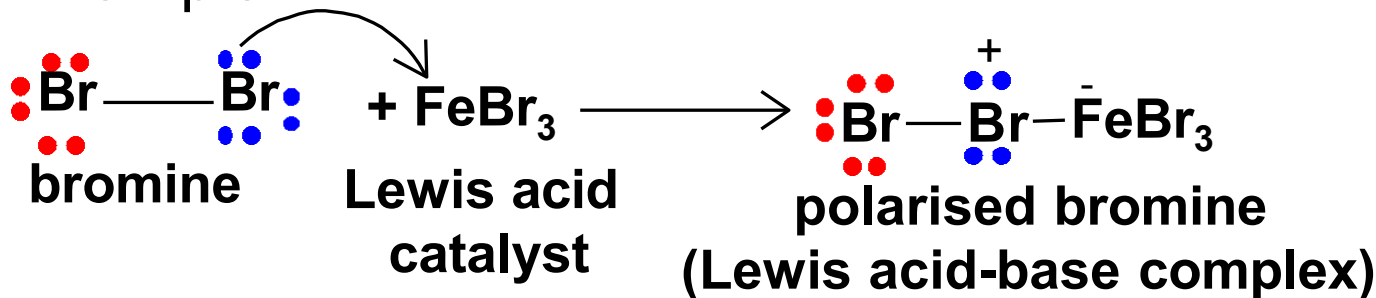
3 Friedel-Crafts Alkylation

4 Friedel-Crafts Acylation



Function of FeX_3 @ AlX_3

1. Both FeX_3 and AlX_3 act as Lewis acid catalyst to form electrophile.
2. Function of Lewis acid catalyst:
 - Polarise the halogen molecule.
 - Remove the halogen from haloalkane molecule to form electrophile by increasing the polarity of C-X bond and produce the carbocation.
3. Example:

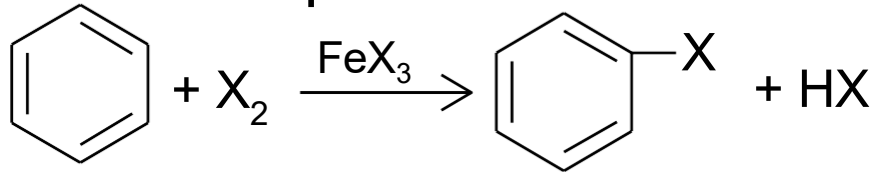


1 Halogenation

Type of reaction : Electrophilic aromatic substitution

Reagents & conditions : $\text{Cl}_2, \text{FeCl}_3$ @ $\text{Br}_2, \text{FeBr}_3$
@ $\text{Cl}_2, \text{AlCl}_3$ @ $\text{Br}_2, \text{AlBr}_3$

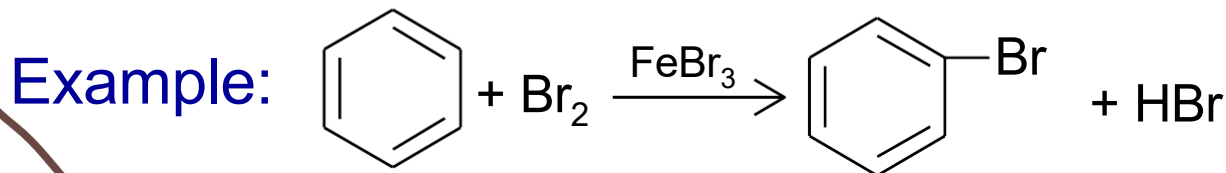
General equation:



Note:

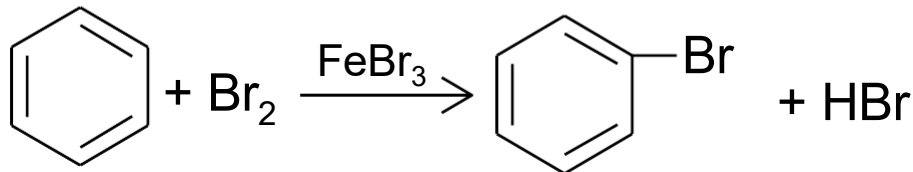
1) Lewis acid used can be FeX_3 or AlX_3

2) $\text{X} = \text{Cl or Br}$



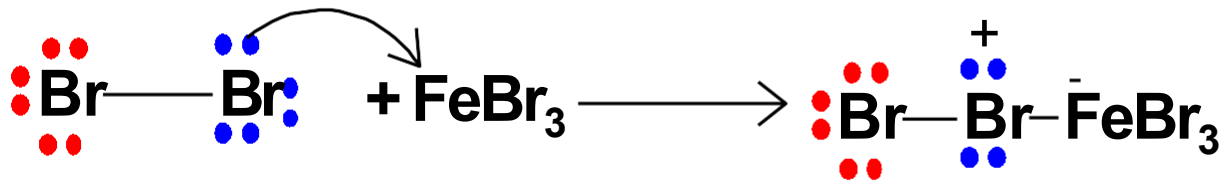
1 Halogenation

Mechanism : Halogenation (Bromination)



Step 1: Formation of electrophile

Bromine combines with FeBr_3 to form a Lewis acid-base complex, FeBr_4^-

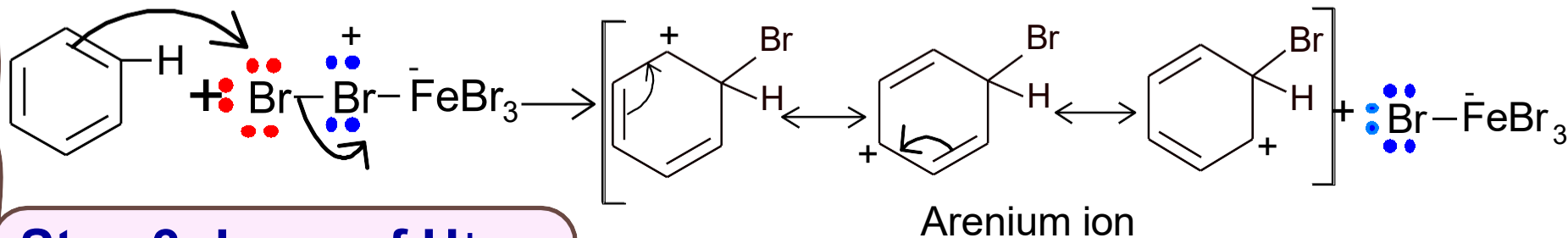


1 Halogenation

1 Electrophilic Aromatic Substitution

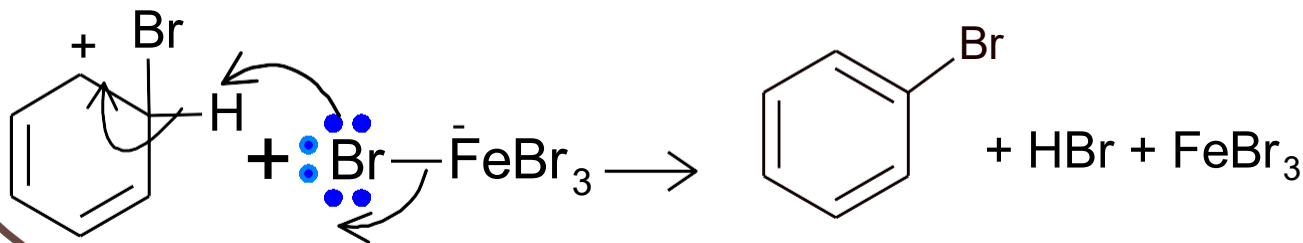
Step 2: Formation of arenium ion

The Lewis acid-base complex attacks benzene to form arenium ion



Step 3: Loss of H^+

A proton is removed from the arenium ion to form bromobenzene

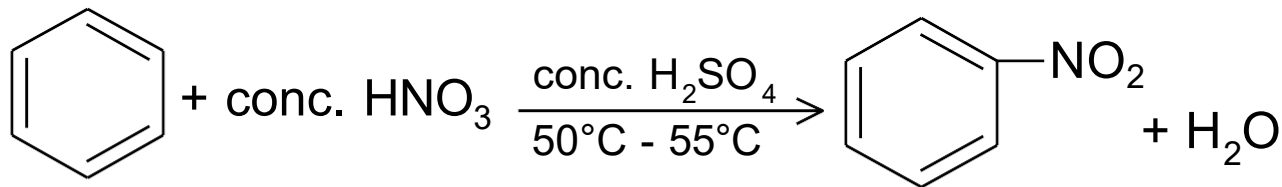


2 Nitration

Type of reaction : Electrophilic aromatic substitution

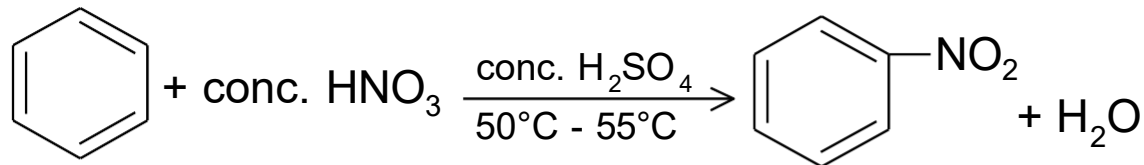
Reagents & conditions : conc HNO_3 , conc H_2SO_4 , $50-55^\circ\text{C}$

Equation :



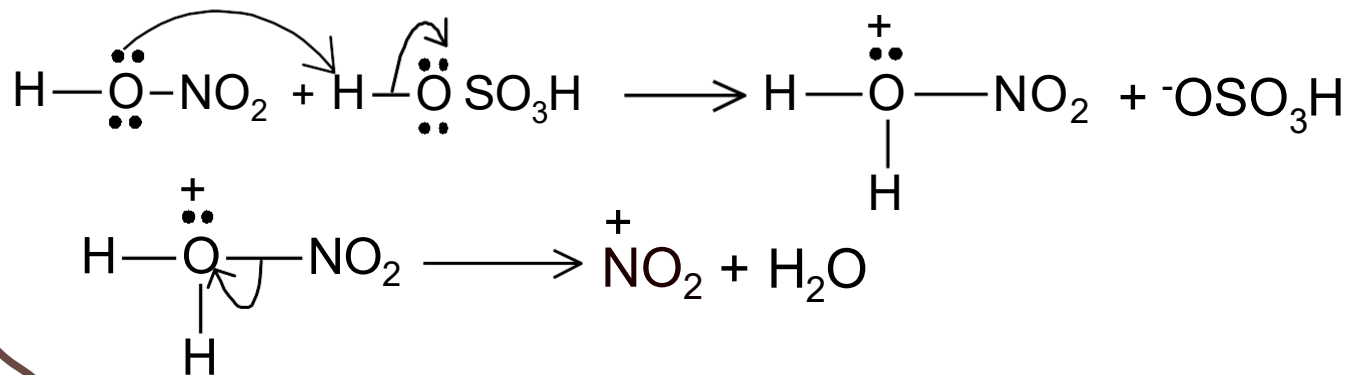
2 Nitration

Mechanism : Nitration



Step 1: Formation of electrophile

Nitric acid accepts proton from sulphuric acid to form nitronium ion



2

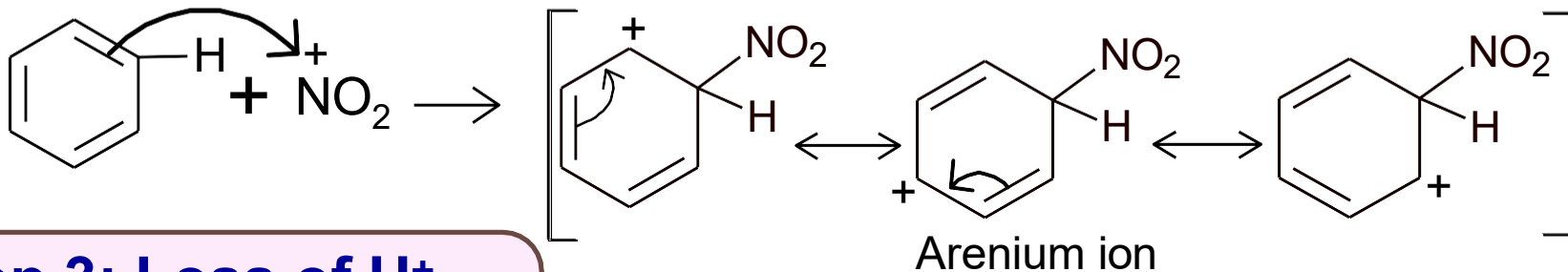
Nitration

1

Electrophilic Aromatic Substitution

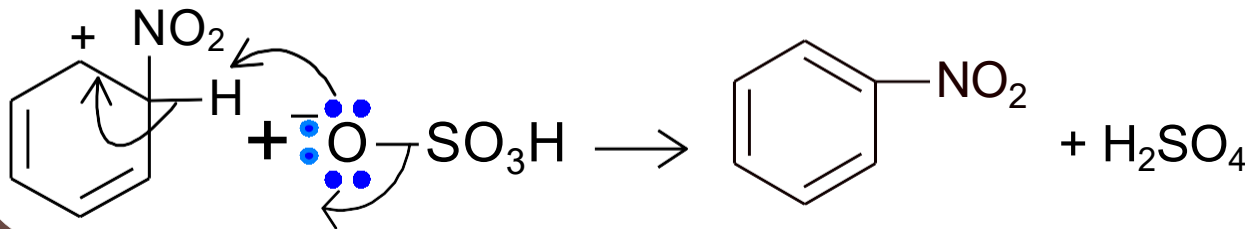
Step 2: Formation of arenium ion

The positive nitronium ion attacks benzene to form arenium ion



Step 3: Loss of H⁺

A proton is removed from the arenium ion to form nitrobenzene



3

Friedel-Crafts Alkylation



French chemist Charles Friedel
(1832 - 1899)

Friedel-Crafts Alkylation

transfer an **alkyl**
group to benzene

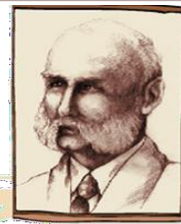


alkyl group, R^+

1 Electrophilic Aromatic Substitution

4

Friedel-Crafts Acylation



American chemist James Mason Crafts
(1839 - 1917)

Friedel-Crafts Acylation

transfer an **acyl** group
to benzene



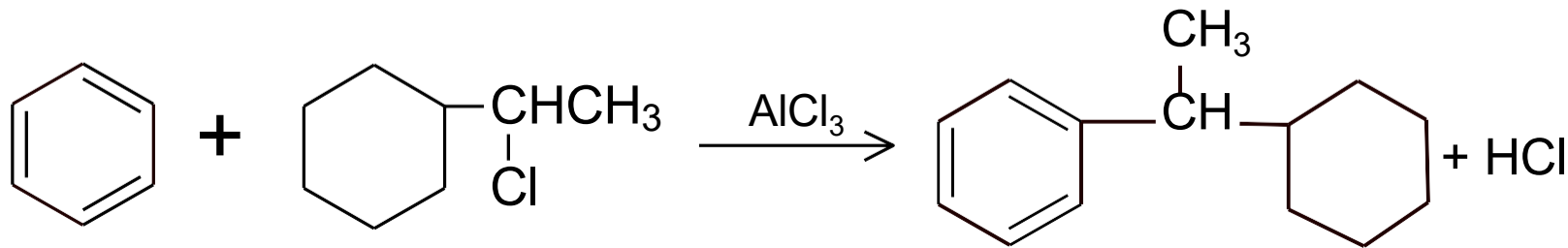
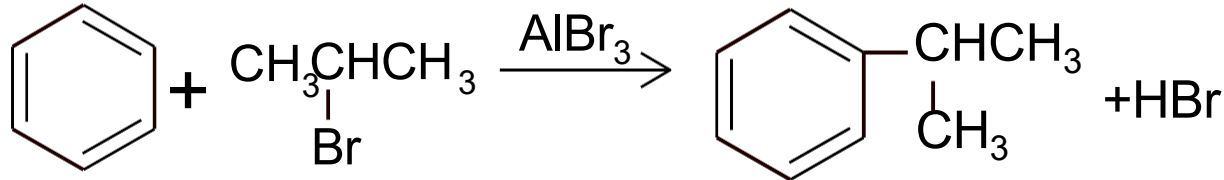
acyl group,
$$R-\overset{\overset{O}{\parallel}}{C}^+$$

3 Friedel-Crafts Alkylation

Type of reaction: Electrophilic aromatic substitution

Reagents & conditions : Haloalkane (RX), Lewis acid catalyst (AlX_3)

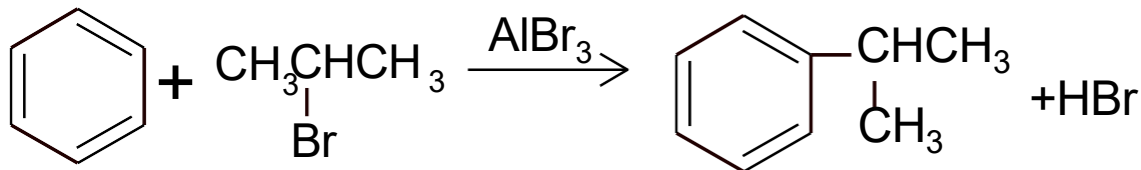
Example:



3 Friedel-Crafts Alkylation

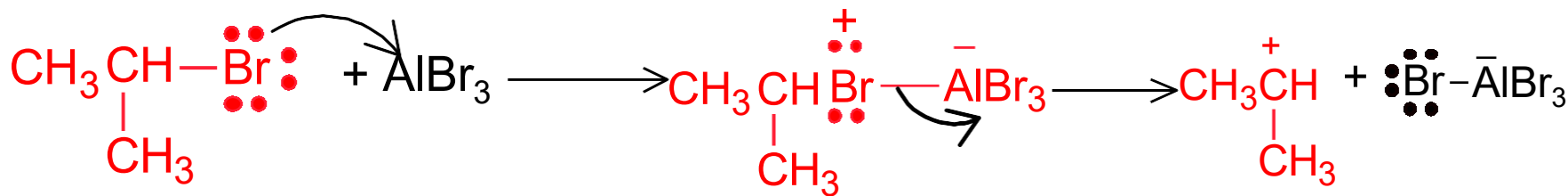
1 Electrophilic Aromatic Substitution

Mechanism : Friedel-Crafts Alkylation



Step 1: Formation of electrophile

The Lewis acid-base complex dissociates to form a carbocation and $-\text{AlBr}_4^-$

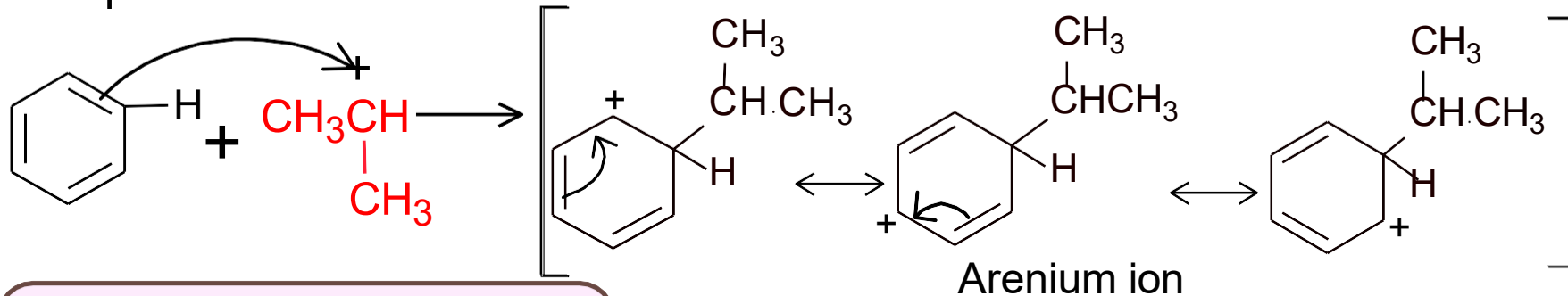


3 Friedel-Crafts Alkylation

1 Electrophilic Aromatic Substitution

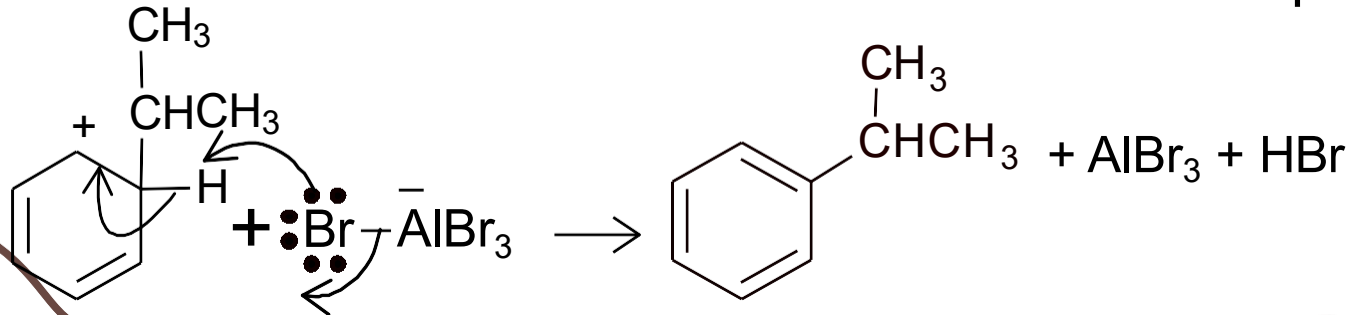
Step 2: Formation of arenium ion

The positive 2° carbocation attacks benzene to form arenium ion



Step 3: Loss of H⁺

A proton is removed from the arenium ion to form isopropylbenzene



3

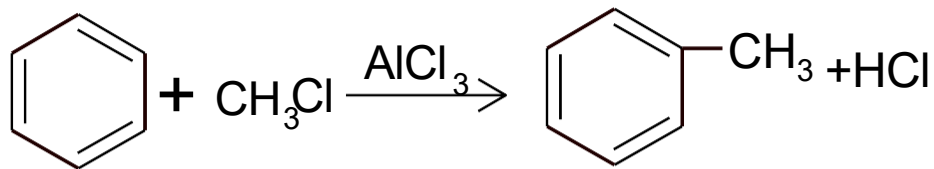
Friedel-Crafts Alkylation

1

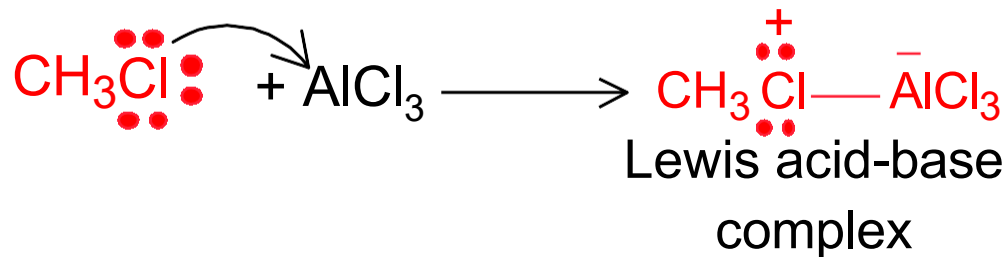
Electrophilic Aromatic Substitution

(Methyl halide : CH_3Cl or CH_3Br or 1o haloalkane)

Mechanism : Friedel-Crafts Alkylation



Step 1: Formation of electrophile

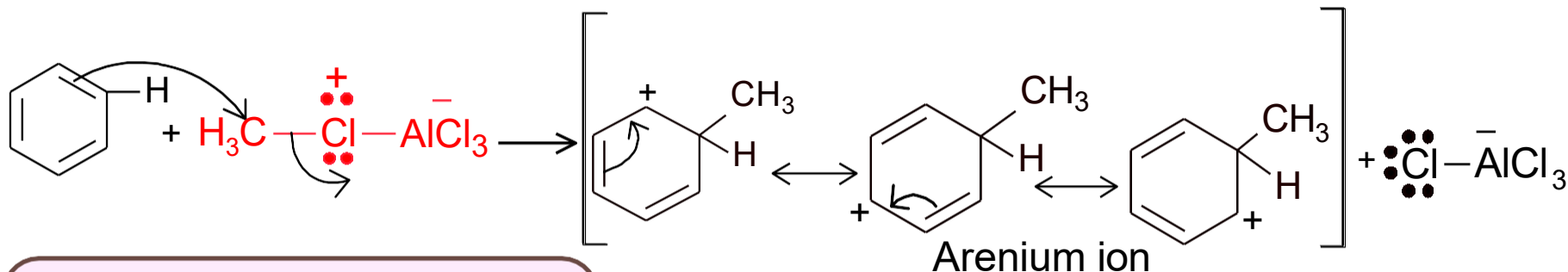


3 Friedel-Crafts Alkylation

1 Electrophilic Aromatic Substitution

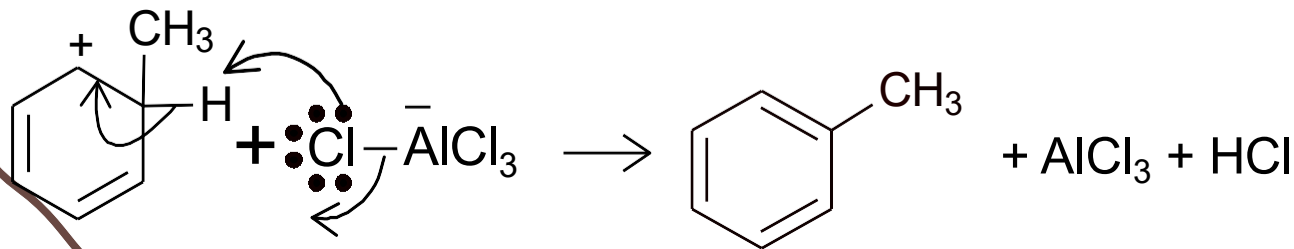
Step 2: Formation of arenium ion

The Lewis acid-base complex attacks benzene to form arenium ion



Step 3: Loss of H^+

A proton is removed from the arenium ion to form isopropylbenzene

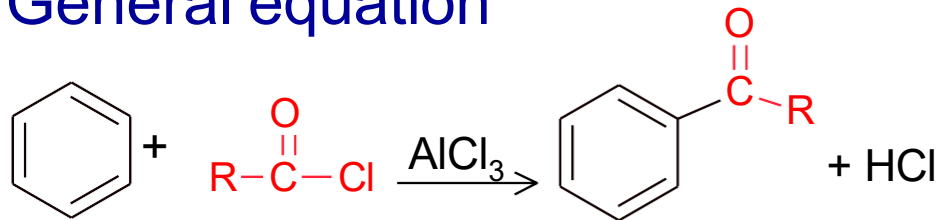


4 Friedel-Crafts Acylation

Type of reaction : Electrophilic aromatic substitution

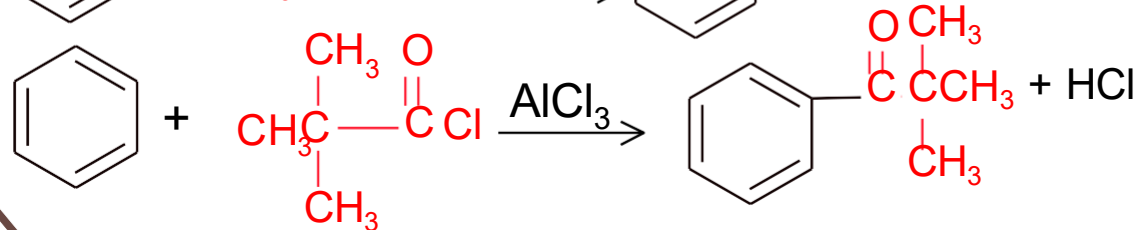
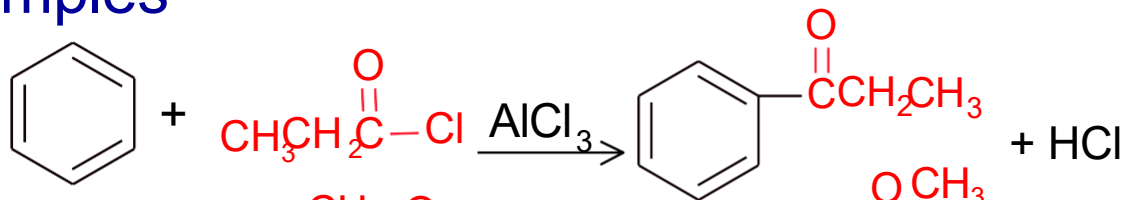
Reagents & conditions : Acyl chloride (RCOCl), Lewis acid catalyst AlCl_3

General equation



Acylation = transfer an **acyl group** to benzene

Examples



SUBSTITUENT EFFECT

1 Electrophilic Aromatic Substitution

Effect of a substituent group on a further substitution

Effect of substituent

Orientation

ortho-para

meta

Reactivity (Reaction Rate)

activating

deactivating

SUBSTITUENT EFFECT

1

Electrophilic Aromatic Substitution

Effect on Reactivity (Reaction Rate)

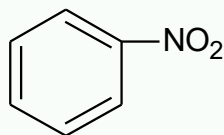
Activating groups

Substituents that activate the ring, making it more reactive than benzene

Meta directors

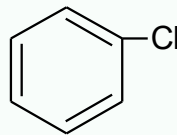
Substituents that deactivate the ring, making it less reactive than benzene

EXAMPLE:
relative rate
of nitration



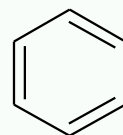
nitrobenzene

6×10^{-8}



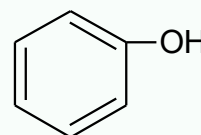
chlorobenzene

0.033



benzene

1



phenol

1000

Increasing reactivity

SUBSTITUENT EFFECT

1

Electrophilic Aromatic Substitution

EFFECT ON ORIENTATION

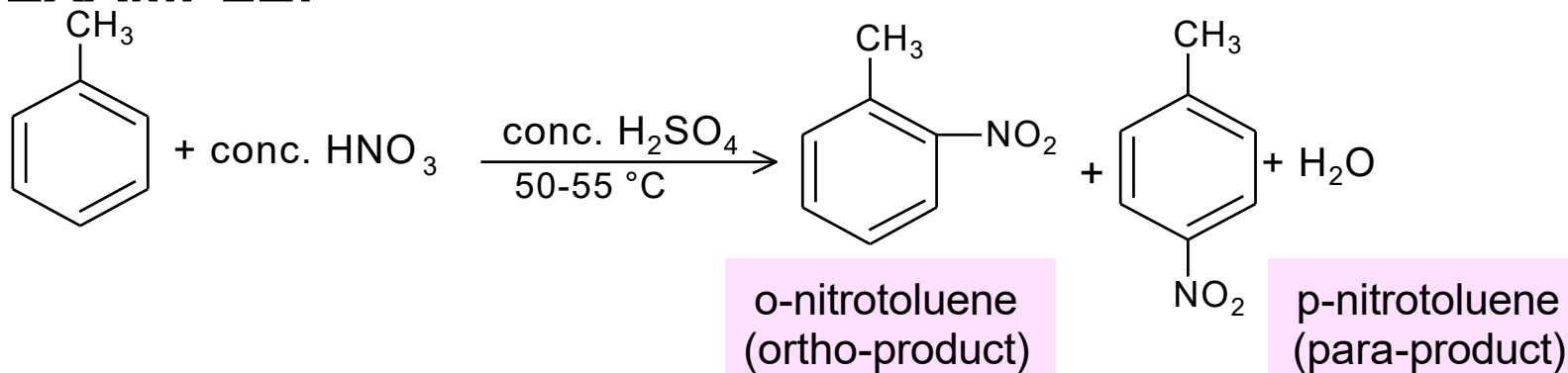
Ortho–para directors

Tend to **direct** the **incoming group** into **ortho** and **para positions**

Deactivating groups

Tend to **direct** the **incoming group** into **meta position**

EXAMPLE:



CLASSIFICATION OF SUBSTITUENTS

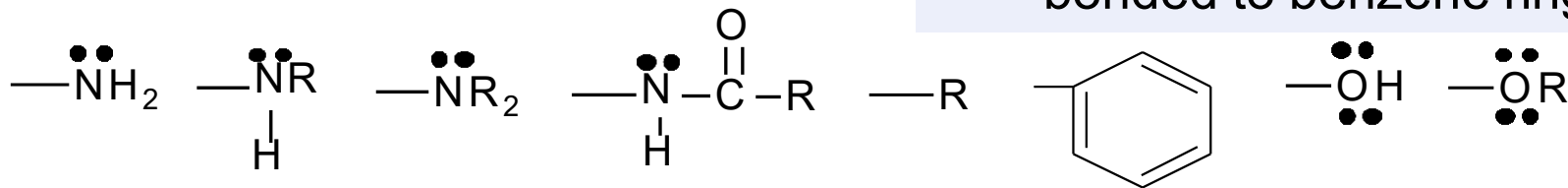
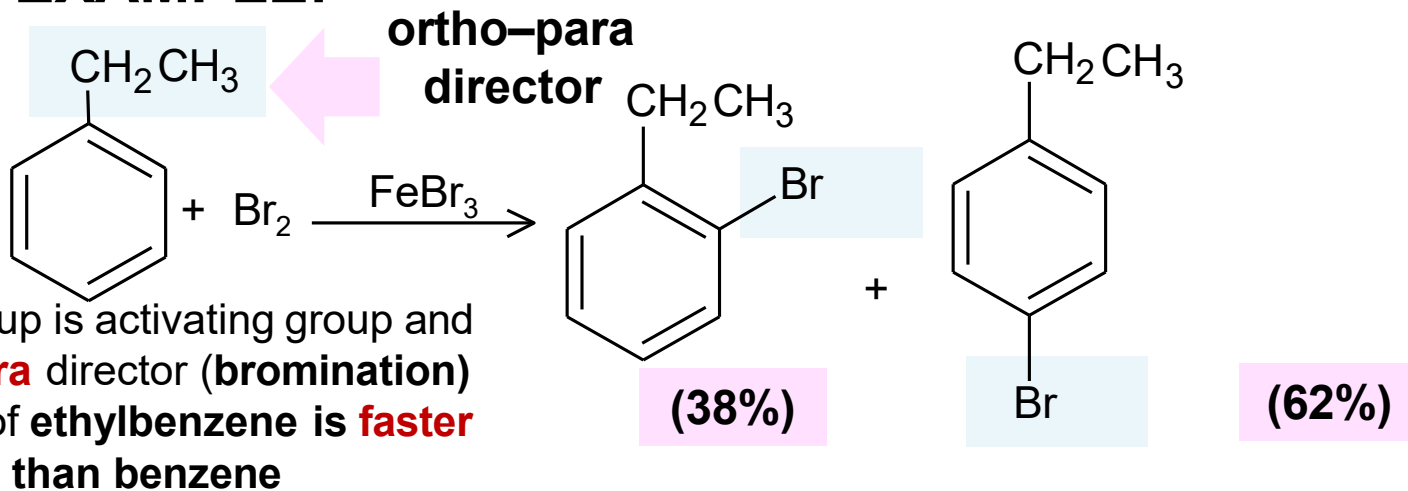
1 Ortho–, para–directing activators

2 Ortho–, para–directing deactivators

3 Meta –directing deactivators

1 Ortho-, para-directing activators

general structure: —R or $\text{—}\ddot{\text{Z}}$ alkyl groups or have nonbonded electron pair on the atom bonded to benzene ring

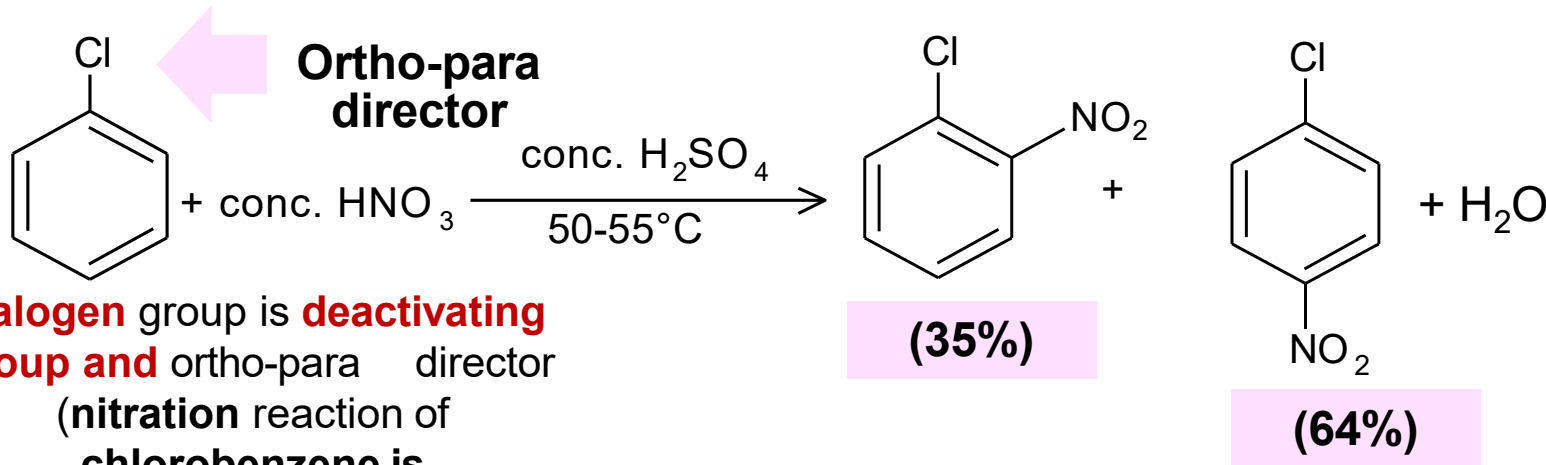
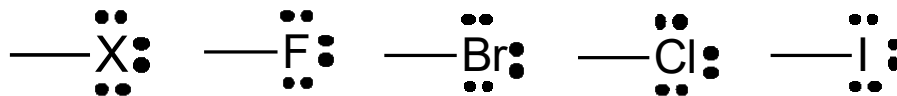
**EXAMPLE:**

2 Ortho-, para-directing deactivators

1 Electrophilic Aromatic Substitution

EXAMPLE:

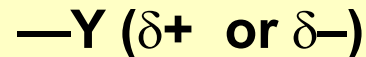
General structure of halogens



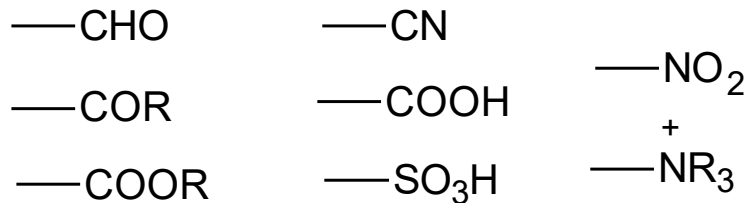
3 Meta-directing deactivators

1 Electrophilic Aromatic Substitution

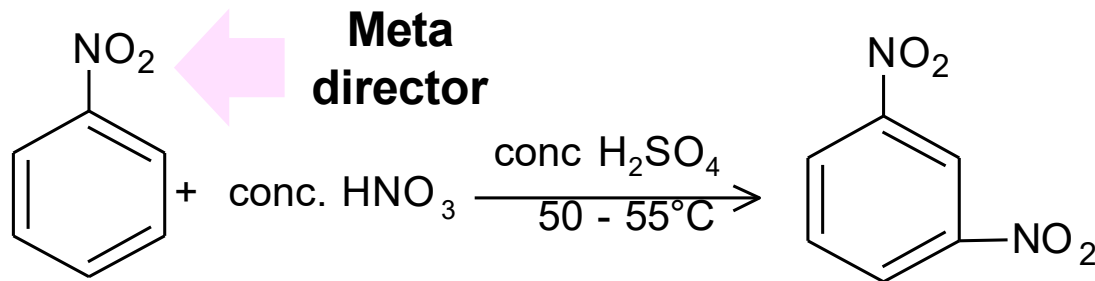
General structures :



have a full or partial positive charge on the atom bonded to benzene ring



EXAMPLE:



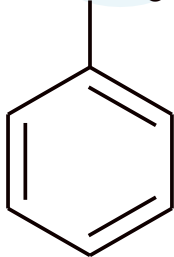
NO_2 group is a deactivating group and **meta director** (nitration reaction of nitrobenzene is **slower** than benzene)

Alkylbenzene

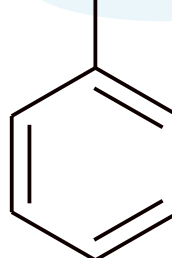
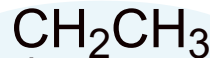
2

Reaction involves substituent attached to benzene ring

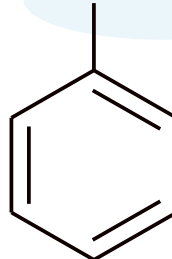
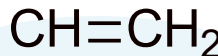
Example :



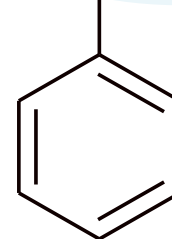
**methylbenzene
(toluene)**



ethylbenzene



phenylethene (styrene or vinylbenzene)



**isopropylbenzene
(cumene)**

Note:

The aliphatic portion of these compounds is called **side chain**.

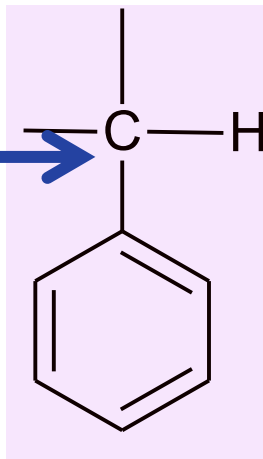
Alkylbenzene

2

Reaction involves substituent attached to benzene ring

Benzylic carbon

Carbon atom which directly bonded to the benzene ring is known as **benzylic carbon**.



Benzylic hydrogen

Hydrogen atom which directly bonded to the benzylic carbon is called **benzylic hydrogen** atom.

Alkylbenzene

2

Reaction involves substituent attached to benzene ring

Reactions of Alkylbenzene

1 Oxidation of alkylbenzene

Reagents/Conditions: KMnO_4 , H^+ , Δ @ $\text{K}_2\text{Cr}_2\text{O}_7$, H^+ , Δ

2 Halogenation (Free Radical Substitution)

Reagents/Conditions: Cl_2 , uv @ Br_2 , uv

Reactions of Alkylbenzene

1 Oxidation of alkylbenzene

Reactant : Alkylbenzene with benzylic hydrogen

Reagent/conditions : Strong oxidising agent

E.g. : $\text{KMnO}_4, \text{H}^+, \Delta$ @ $\text{K}_2\text{Cr}_2\text{O}_7, \text{H}^+, \Delta$ @ $\text{Na}_2\text{Cr}_2\text{O}_7, \text{H}^+, \Delta$

In **oxidation of alkylbenzene**,

- **Alkyl group** (with **benzylic H**) regardless its chain length, is converted to **–COOH**.
- **alkyl group** (without benzylic H) can't be oxidised.

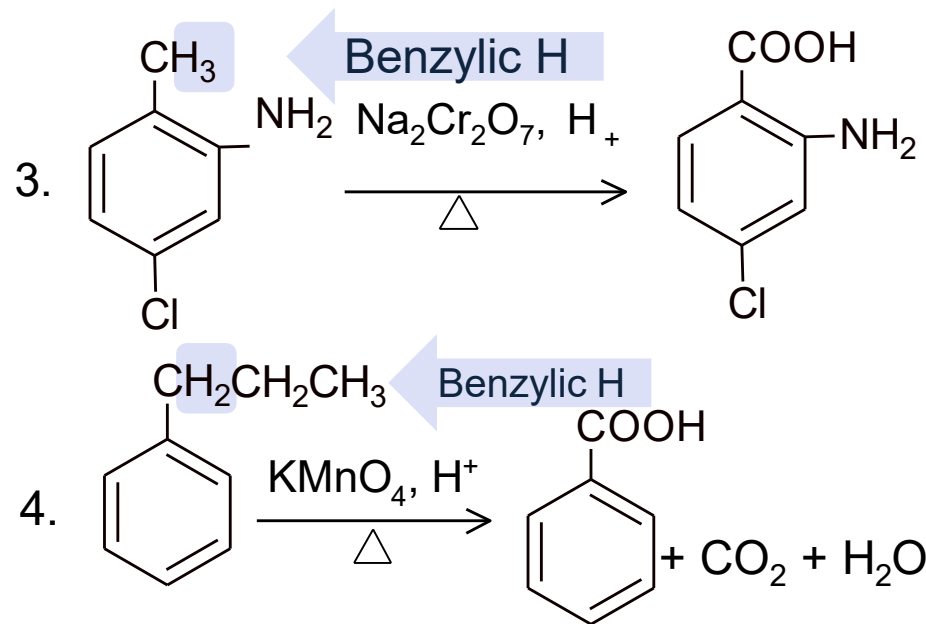
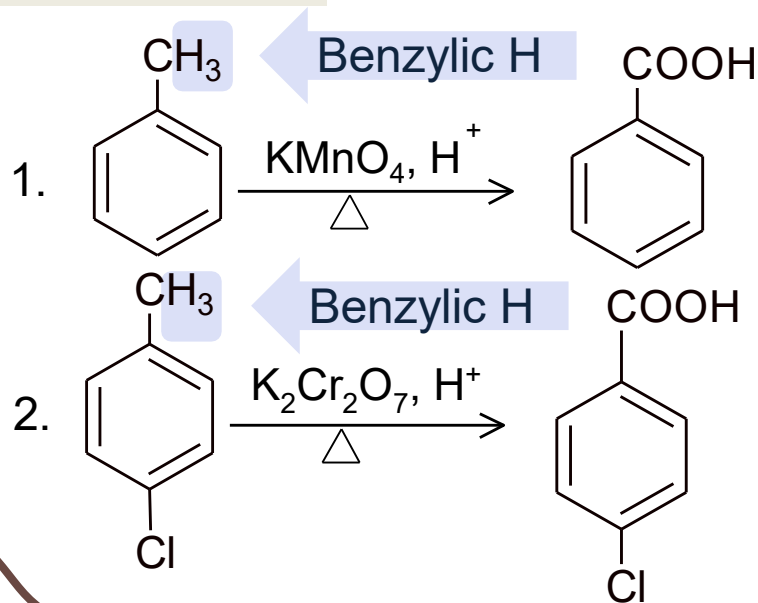
2

Reaction involves substituent attached to benzene ring

Reactions of Alkylbenzene

1 Oxidation of alkylbenzene

Example:



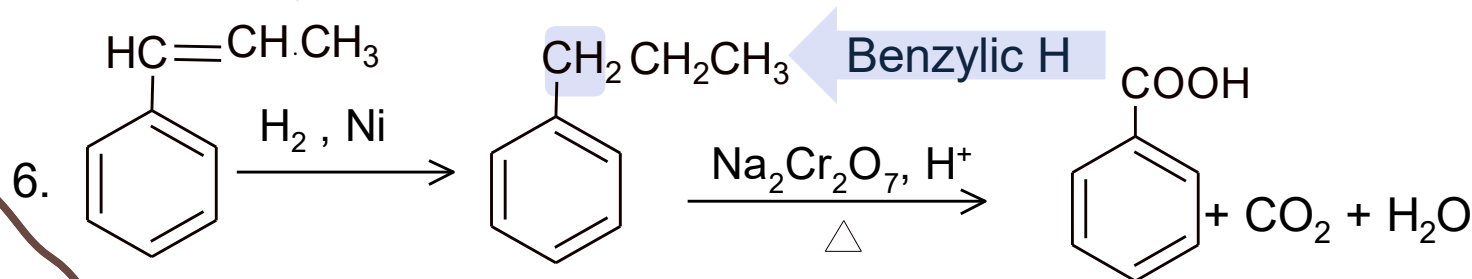
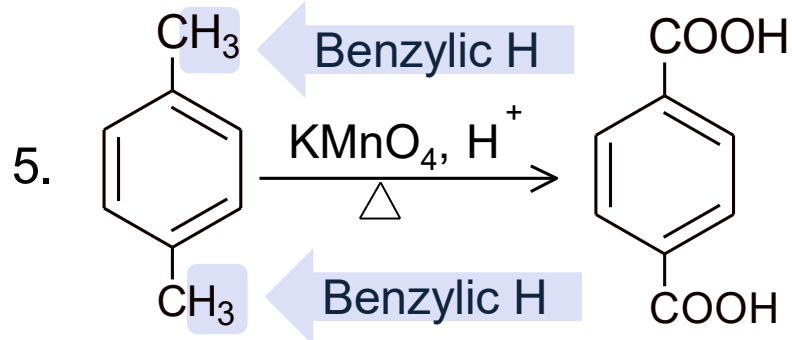
2

Reaction involves substituent attached to benzene ring

Reactions of Alkylbenzene

1 Oxidation of alkylbenzene

Example:



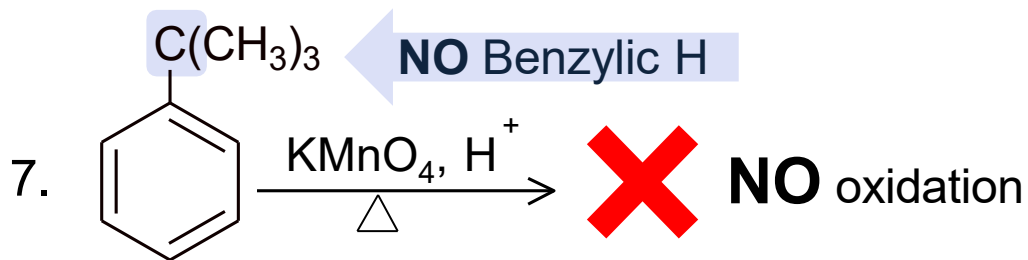
2

Reaction involves substituent attached to benzene ring

Reactions of Alkylbenzene

1 Oxidation of alkylbenzene

Example:



KEEP IN MIND !!!!

Compounds **without a benzylic H** are **can't be oxidised**.

Reactions of Alkylbenzene

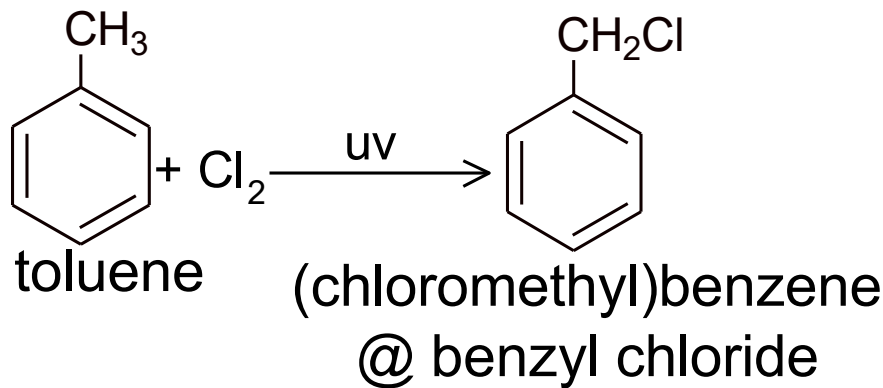
2 Halogenation (Free Radical Substitution)

- Halogenation of alkylbenzene can take place at the side chain
- Involves free radical substitution.
- Free radical substitution takes place at:
 - ✓ high temperature or
 - ✓ in the presence of uv light

Reactions of Alkylbenzene

2 Halogenation (Free Radical Substitution)

Example:



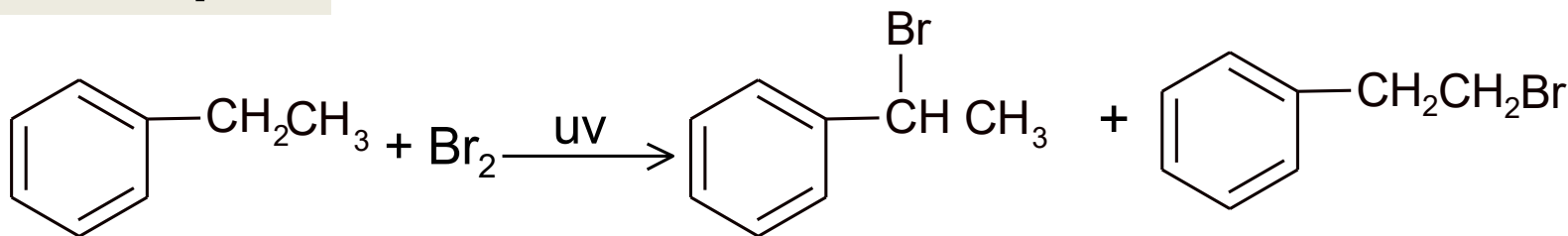
Note :

Cl or Br replaces H
atom of the alkyl group.

Reactions of Alkylbenzene

2 Halogenation (Free Radical Substitution)

Example:



Major product

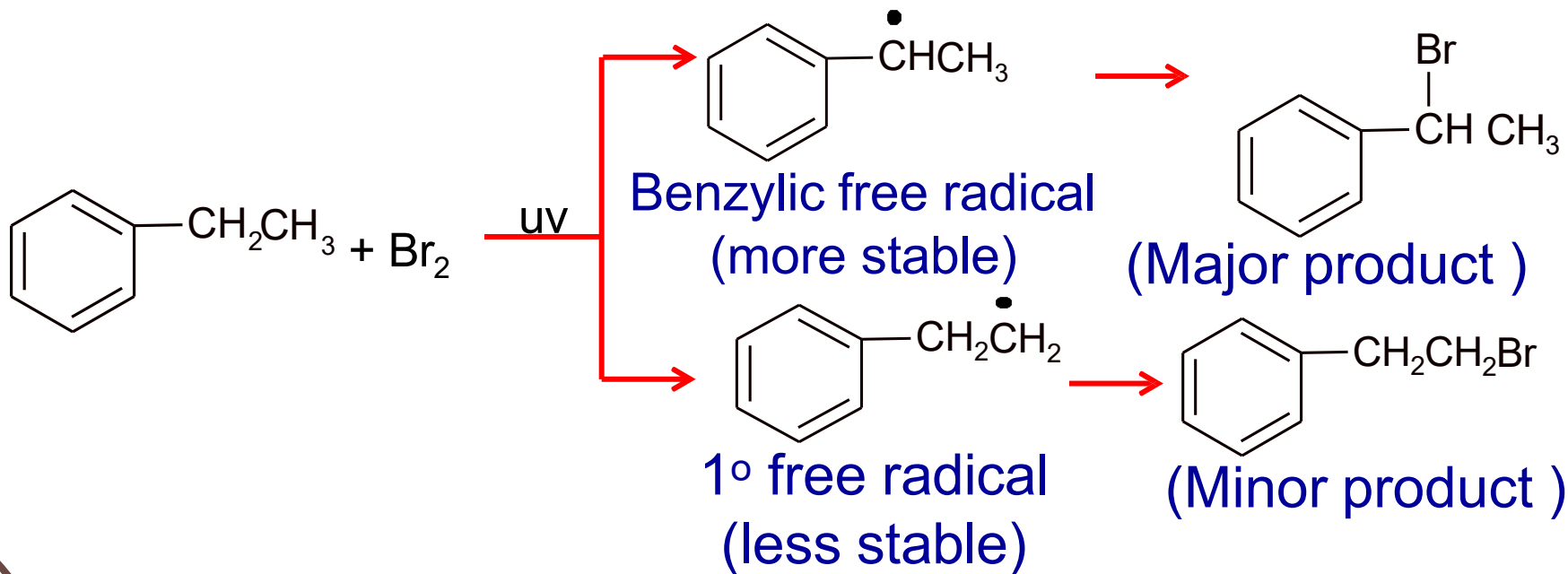
Major product because it is formed from **more stable benzylic free radical**

Minor product

Minor product because it is formed from **less stable 1° free radical**.

Reactions of Alkylbenzene

2 Halogenation (Free Radical Substitution)

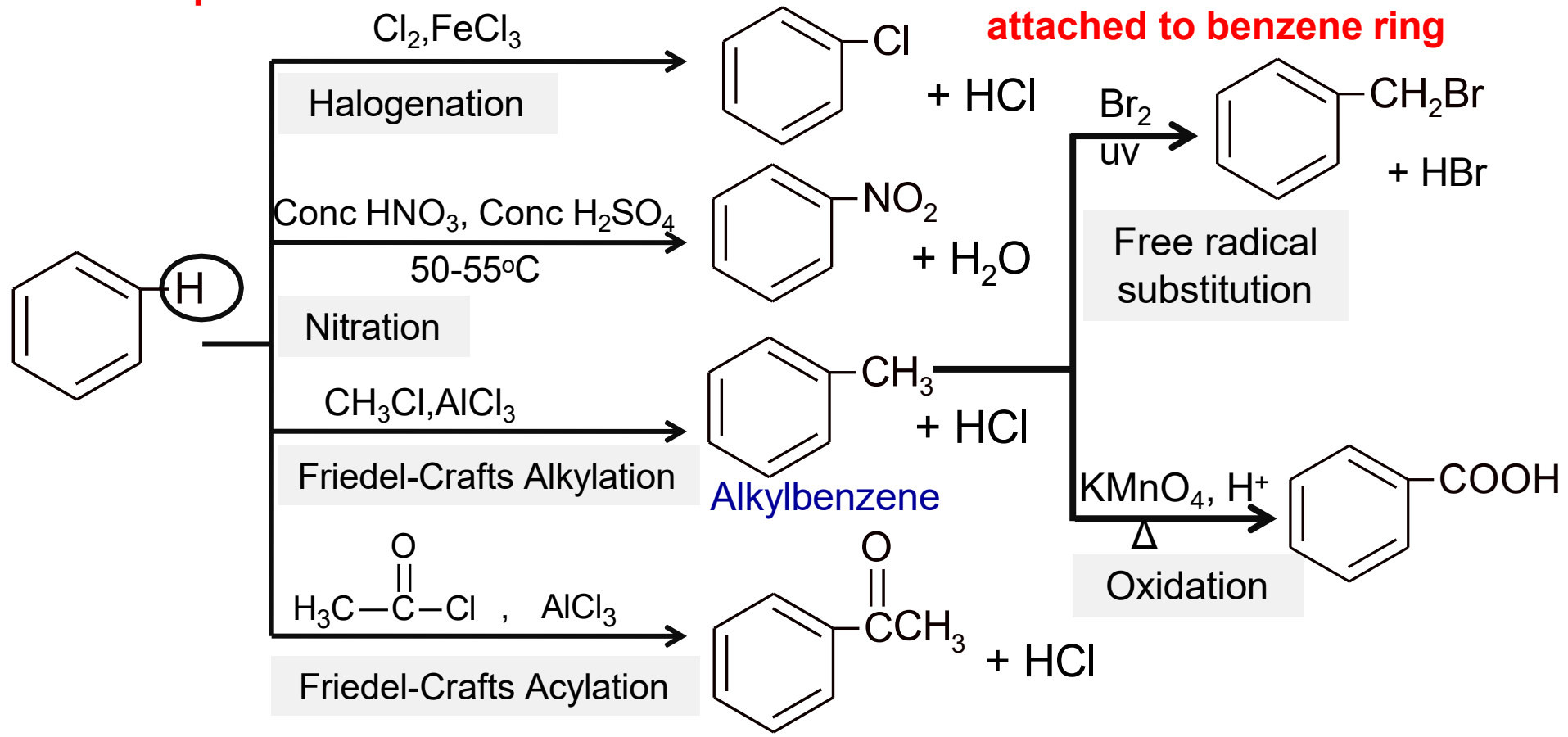


Benzylic free radicals are even more stable than 3° free radicals!!

SUMMARY OF REACTIONS INVOLVING BENZENE

Electrophilic Aromatic Substitution

Reaction involves substituent attached to benzene ring



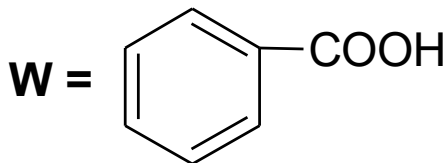
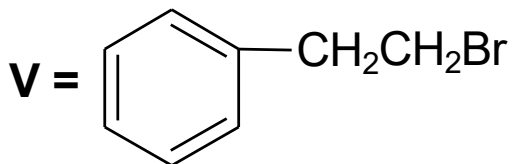
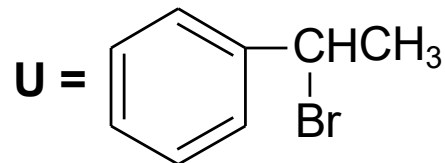
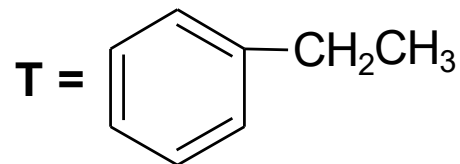
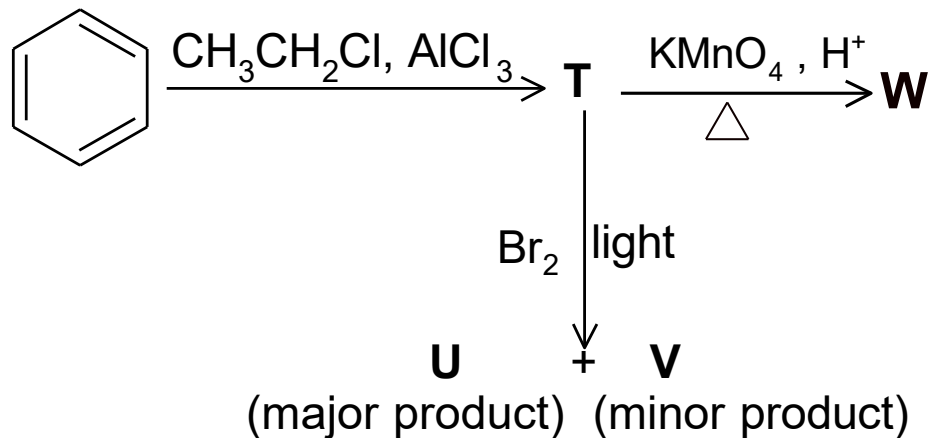
EXAMPLE 1 (PYQ 2011/2012)

Compound **T** is formed from the reaction of benzene with chloroethane in the presence of AlCl_3 . Bromination of **T** in light gives monobrominated compounds **U** (**major** product) and **V**. Treatment of **T** in hot acidified KMnO_4 produces **W**. Draw the structures of compounds **T**, **U**, **V** and **W**. Name the reaction for the formation of **T** and write the mechanism.

EXAMPLE 1 (PYQ 2011/2012)

59

ANSWER

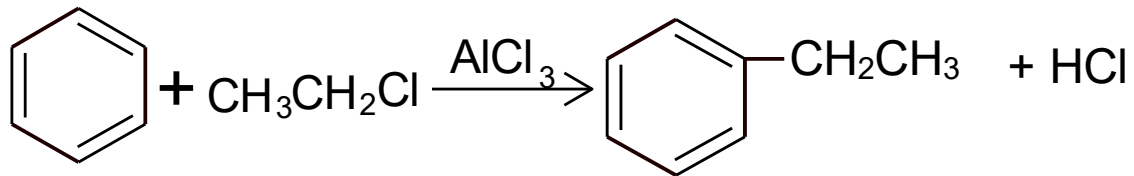


EXAMPLE 1 (PYQ 2011/2012)

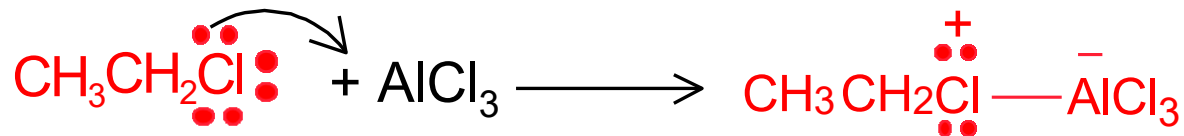
60

ANSWER

Name of reaction : Friedel-Crafts Alkylation



Step 1: Formation of electrophile



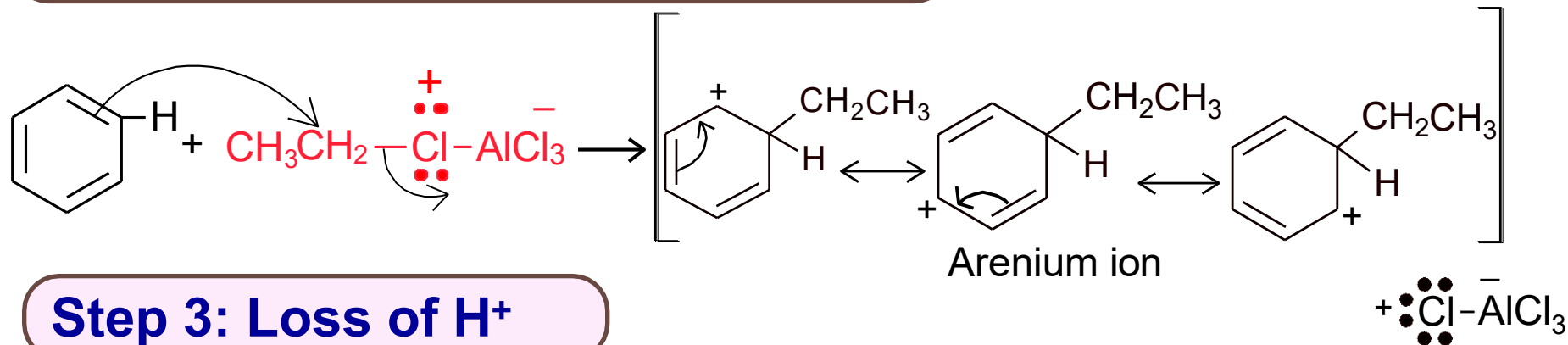
Lewis acid-base complex

EXAMPLE 1 (PYQ 2011/2012)

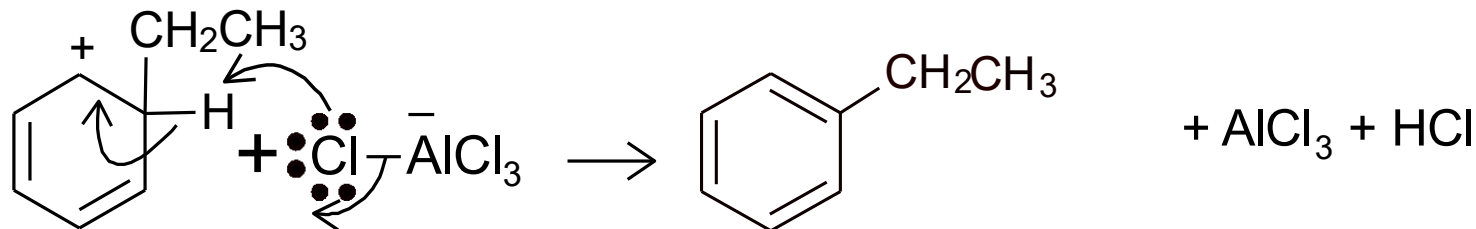
61

ANSWER

Step 2: Formation of arenium ion



Step 3: Loss of H^+



EXAMPLE 2

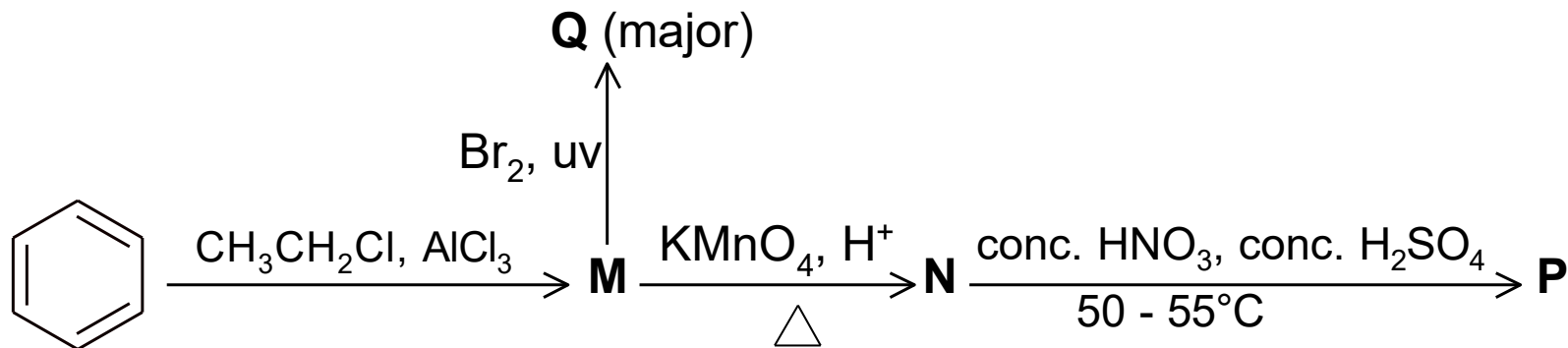
Benzene reacts with chloroethane in the presence of aluminium chloride to form **M**. Oxidation of **M** forming **N** that turns blue litmus paper to red. **N** undergoes nitration to form **P**. Bromination of **M** in the presence of sunlight forming **Q** as major product.

- i. Name the reaction involves in formation of **M**.
- ii. Draw the structural formulae of compounds **M**, **N**, **P** and **Q**.
- iii. Give the IUPAC name for compound **P**.

EXAMPLE 2

63

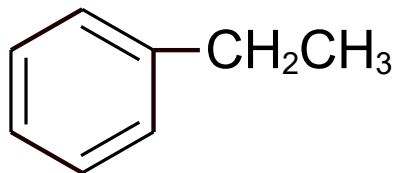
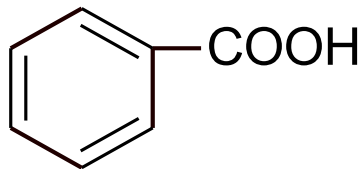
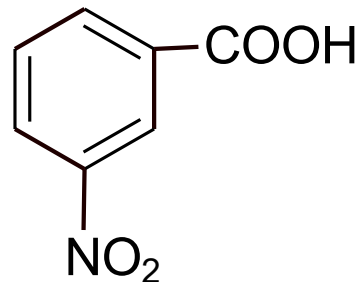
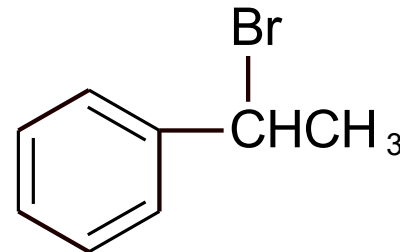
ANSWER



- i. Name of reaction: Friedel – Crafts alkylation

ANSWER

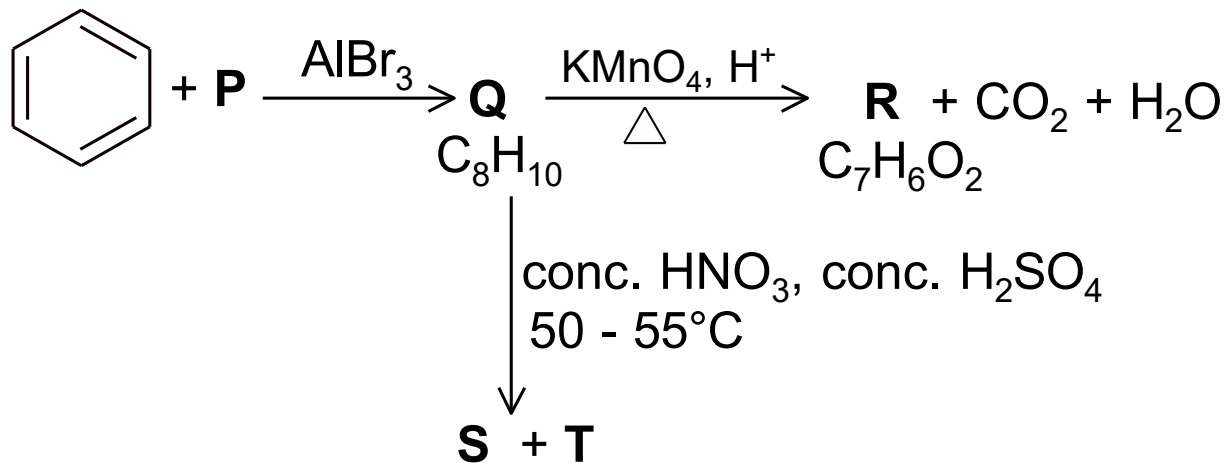
ii.

**M****N****P****Q**

iii. **P** = 3-nitrobenzoic acid @ m-nitrobenzoic acid

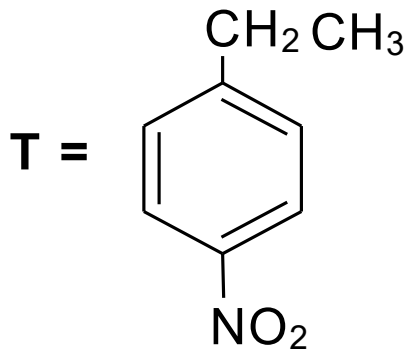
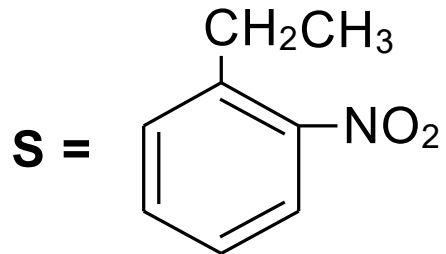
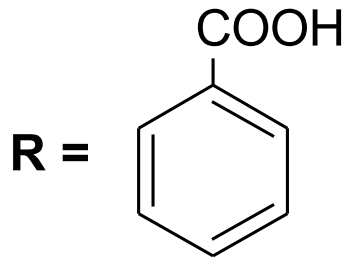
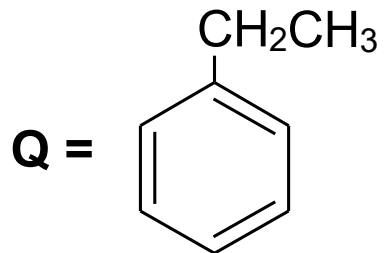
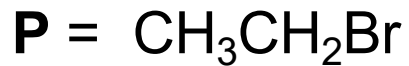
EXAMPLE 3

Draw the structural formulae of **P**, **Q**, **R**, **S** and **T** in the reaction scheme below.



EXAMPLE 3

ANSWER

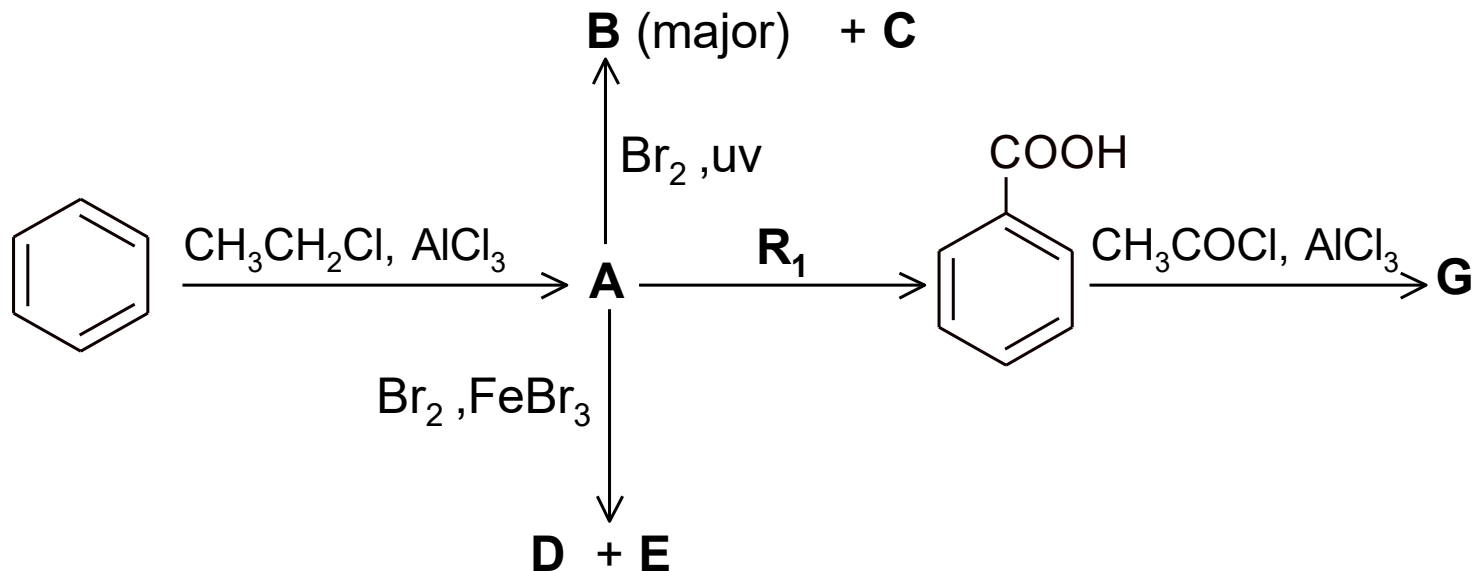


S and **T** are interchangeable

EXAMPLE 4

Base on the reaction scheme below:

- Draw structural formulae of compounds **A** to **E**
- Suggest suitable reagent and conditions for **R₁**

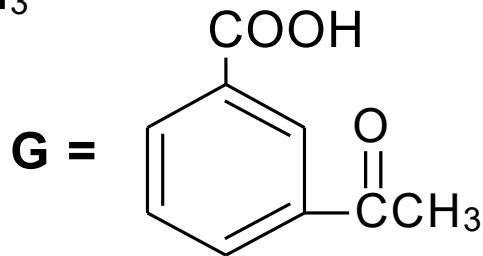
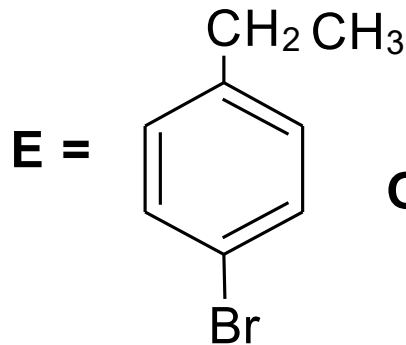
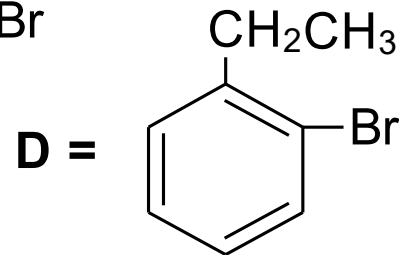
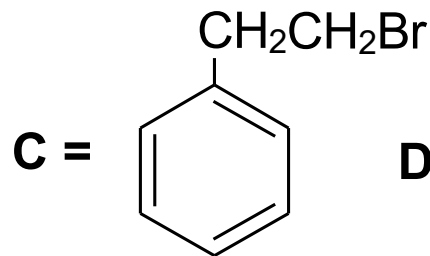
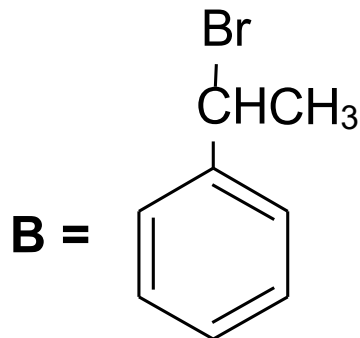
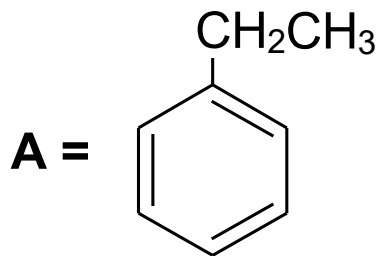


EXAMPLE 4

68

ANSWER

i.

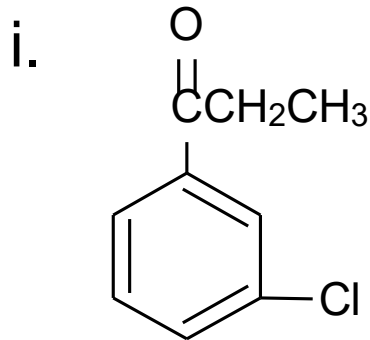


D and **E** are interchangeable

ii. $R_1 = \text{KMnO}_4, \text{H}^+, \Delta @ \text{K}_2\text{Cr}_2\text{O}_7, \text{H}^+, \Delta @ \text{Na}_2\text{Cr}_2\text{O}_7, \text{H}^+, \Delta$

EXAMPLE 5

Show how each of the following compounds can be prepared from benzene:



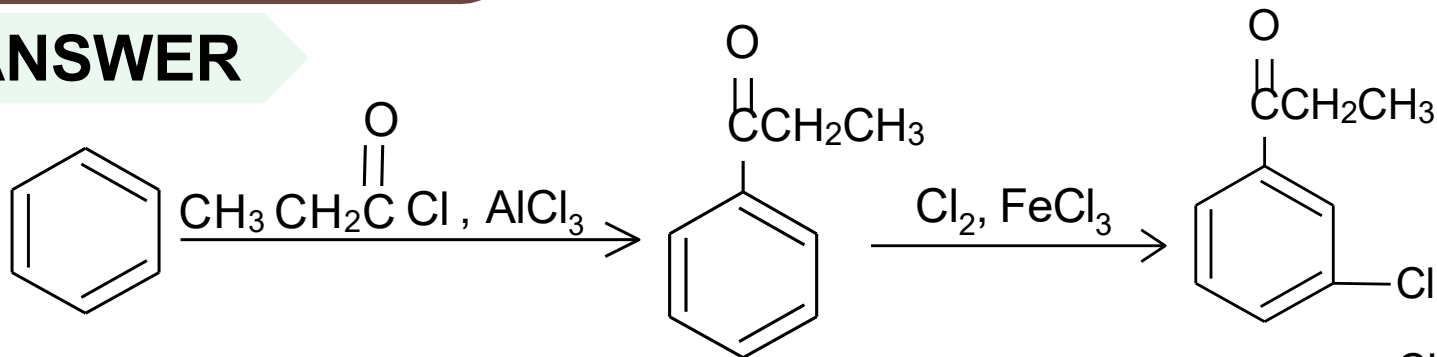
ii. p-nitrobenzoic acid

EXAMPLE 5

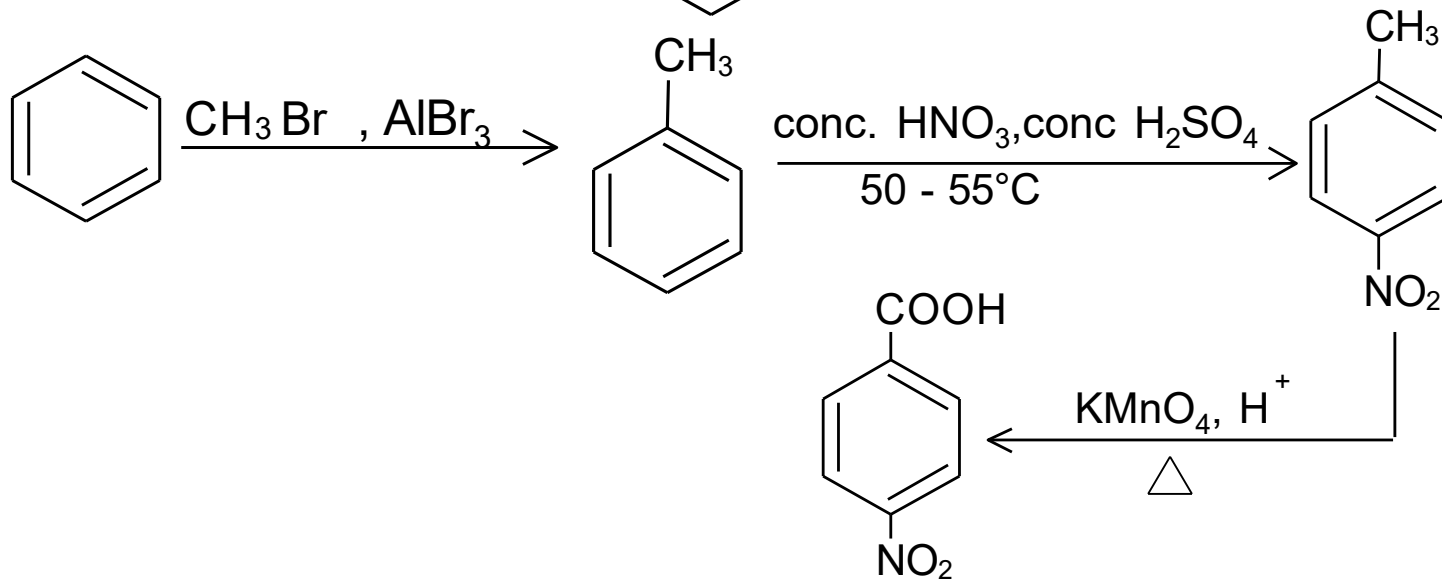
70

ANSWER

i.



ii.



The slide features abstract geometric designs in the corners. The top-left corner is filled with overlapping triangles in shades of blue, green, and red. The bottom-right corner contains a cluster of overlapping triangles in various shades of gray. A small vertical bar with four colored segments (blue, green, red, and orange) is positioned to the left of the main text.

THE END